Genuine Multipartite Entanglement Analysis — Molecular System

MOLECULE

Benzene (C6H6)
Basis set   : cc-pVDZ
Charge   : 0
Multiplicity : 1
Total e-   : 42
n-body order : 3

GROUND STATE ENERGIES

E0 (Ha) : -230.75355805
E1 (Ha) : -230.57826157
Gap (Ha) : 0.17529648
HF (Ha) : -230.71878874

CONVEX-ROOF METHODS

Methods run   : gcg
Restarts   : 8

C_LB : coherence witness bound
C_expo: unitary opt (exp map)
C_stf : Stiefel manifold (Alg. B)
C_UB : eigendecomposition bound

ACTIVE SPACE CAS(6,6)

Frozen core orbs : 18
Frozen e-   : 36
Active e-   : 6
Active orbs   : 6
Spin-orbitals   : 12
Hilbert dim   : 400

SPIN PROPERTIES

<S2>   : 0.000000
S   : 0.0000
<Sz>   : 0.000000
Multiplicity : 1
Spin state   : Singlet (S=0)

GME CRITERIA

C_GME = sqrt(2 * min_A (1 - Tr(rho_A^2)))

Convex roof (mixed):
  C_GME(rho) = min Sum p_i C_GME(|psi_i>)

Bounds ordering:
  C_LB <= C_GME <= C_UB

DOUBLE OCCUPANCY

<D> total : 2.961506
<D> / site : 0.493584

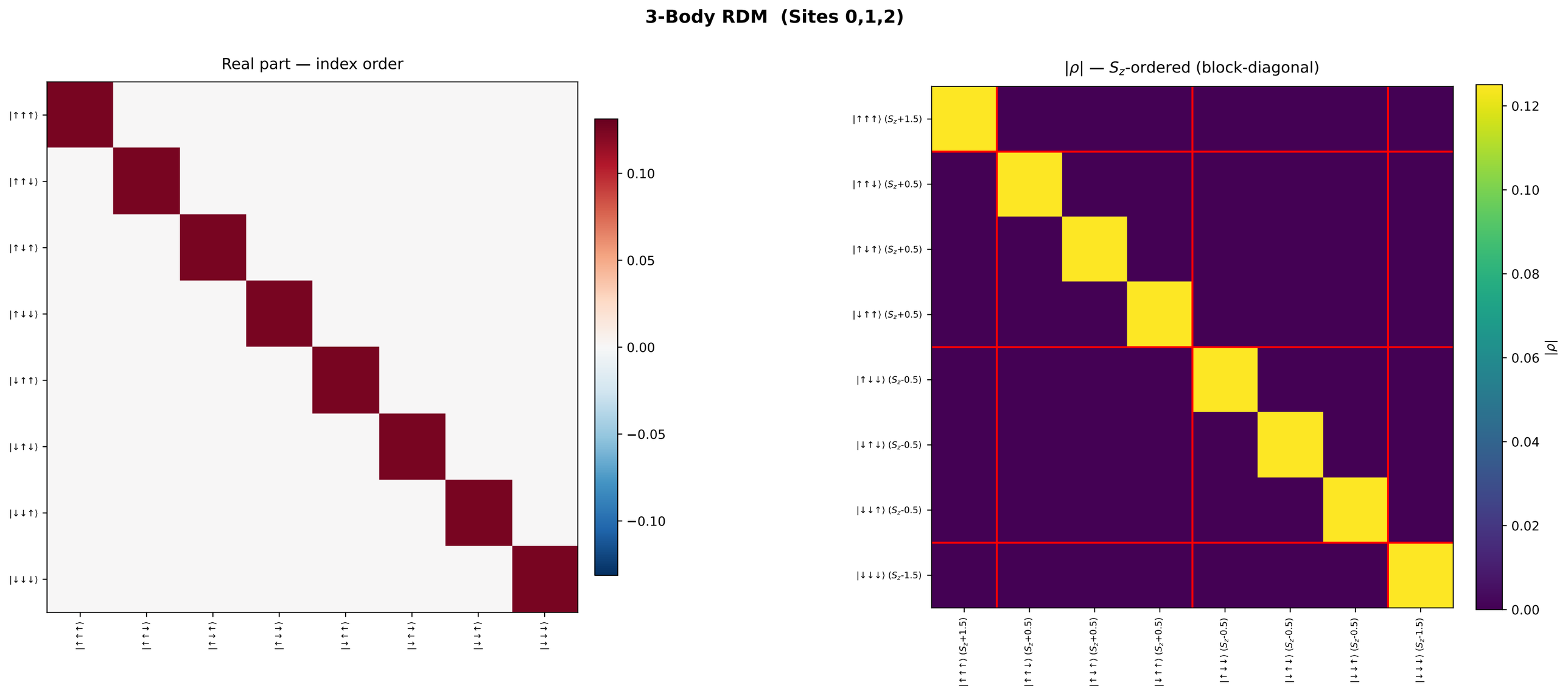
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Body order : 3
Clusters   : 41

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Summary — 3-Body RDMs (Benzene, CAS(6,6))

Orbitals	Purity	C_GME (pure)	C_LB	C_gcg	C_stiefel	C_UB	State
0,1,2	0.125000	—	—	—	—	—	STRONGLY MIXED
0,1,3	0.148891	—	—	—	—	—	STRONGLY MIXED
0,1,4	0.125559	—	—	—	—	—	STRONGLY MIXED
0,1,5	0.125067	—	—	—	—	—	STRONGLY MIXED
0,2,3	0.125559	—	—	—	—	—	STRONGLY MIXED
0,2,4	0.148904	—	—	—	—	—	STRONGLY MIXED
0,2,5	0.125065	—	—	—	—	—	STRONGLY MIXED
0,3,4	0.125949	—	—	—	—	—	STRONGLY MIXED
0,3,5	0.150430	—	—	—	—	—	STRONGLY MIXED
0,4,5	0.150975	—	—	—	—	—	STRONGLY MIXED
1,2,3	0.496207	—	—	—	—	—	MIXED
1,2,4	0.486887	—	—	—	—	—	MIXED
1,2,5	0.125000	—	—	—	—	—	STRONGLY MIXED
1,3,4	0.499664	—	—	—	—	—	MIXED
1,3,5	0.145172	—	—	—	—	—	STRONGLY MIXED
1,4,5	0.125941	—	—	—	—	—	STRONGLY MIXED
2,3,4	0.499782	—	—	—	—	—	MIXED
2,3,5	0.125967	—	—	—	—	—	STRONGLY MIXED
2,4,5	0.144255	—	—	—	—	—	STRONGLY MIXED
3,4,5	0.125368	—	—	—	—	—	STRONGLY MIXED


3-Body RDM (Sites 0,1,2)
Real part — index order
|↑↑↑⟩
|↑↑↓⟩
|↑↓↑⟩
|↑↓↓⟩
|↓↑↑⟩
|↓↑↓⟩
|↓↓↑⟩
|↓↓↓⟩
0.10
0.05
0.00
−0.05
−0.10
|ρ| — S_z-ordered (block-diagonal)
|↑↑↑⟩ (S_z+1.5)
|↑↑↓⟩ (S_z+0.5)
|↑↓↑⟩ (S_z+0.5)
|↓↑↑⟩ (S_z+0.5)
|↑↓↓⟩ (S_z-0.5)
|↓↑↓⟩ (S_z-0.5)
|↓↓↑⟩ (S_z-0.5)
|↓↓↓⟩ (S_z-1.5)
0.12
0.10
0.08
0.06
0.04
0.02
0.00
|ρ|
Trace=1.000000 Purity=0.125000 Entropy=2.079441 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)

Analysis Summary - 3-Body RDM (Sites 0,1,2)

RDM Metrics
==========

Trace(rho)	: 1.0000000000
Purity Tr(rho^2)	: 0.1250000348
Von Neumann entropy	: 2.0794414025
Eigenvalue min	: 1.248858e-01
Eigenvalue max	: 1.250381e-01
Imaginary part	: < 1e-6 (not shown)

GME Analysis
==========

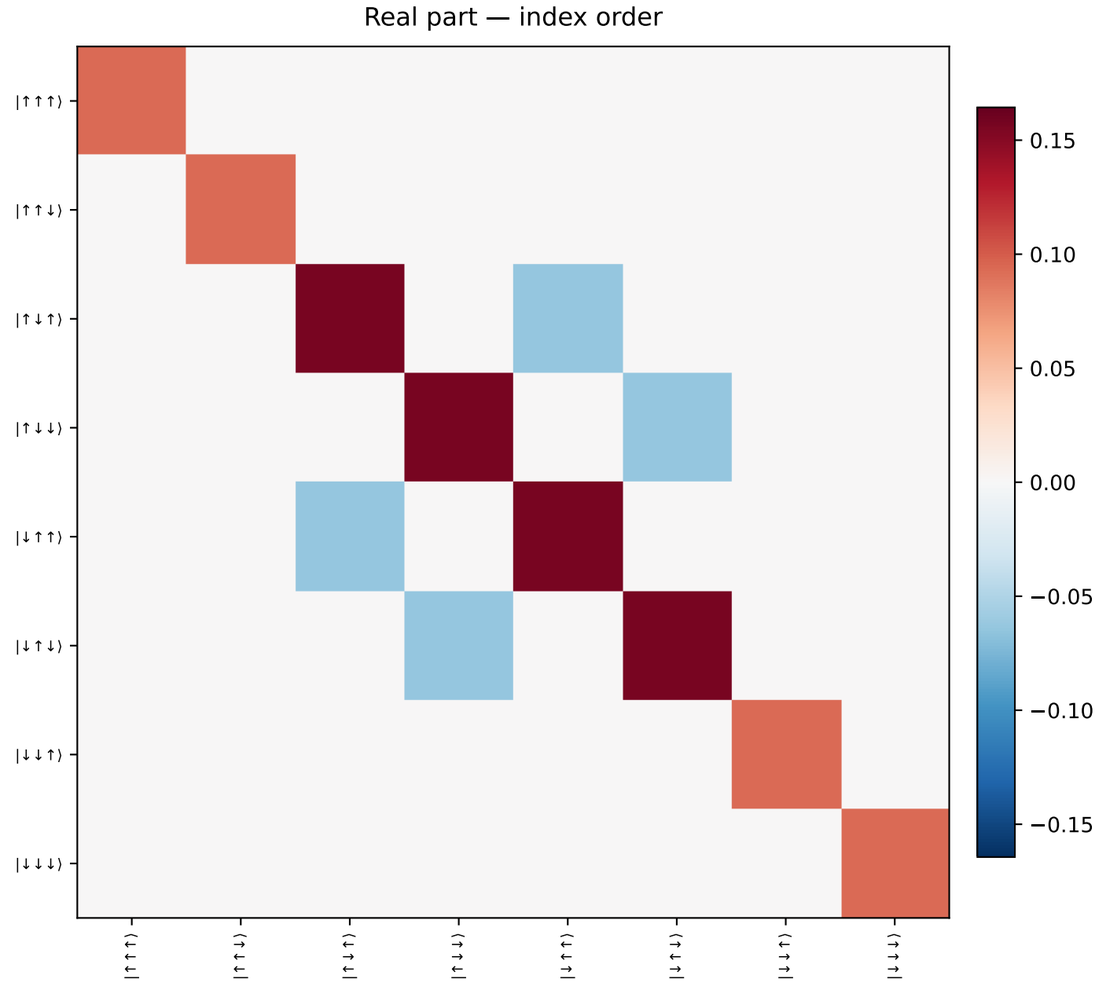
gamma = Tr(rho^2) = 0.12500003

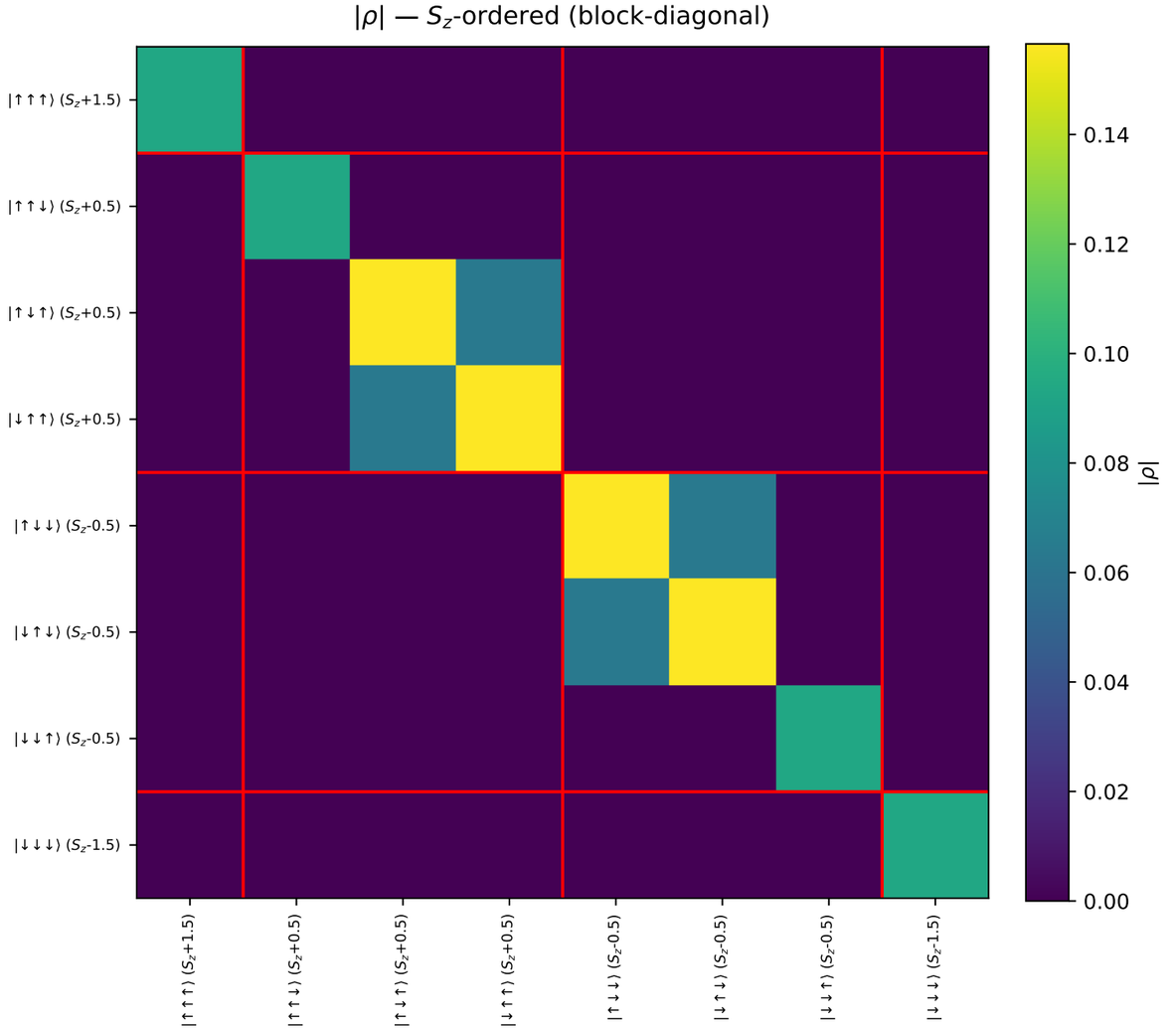
A	B	Tr(rho_A^2)

STATUS: MIXED STATE

Tr(rho^2) < 1. Pure-state C_GME not applicable. See convex-roof result below.

3-Body RDM (Sites 0,1,3)





Trace=1.000000 Purity=0.148891 Entropy=1.994894 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 ($\Rightarrow S_z$ conserved)

Analysis Summary - 3-Body RDM (Sites 0,1,3)

RDM Metrics
==========

Trace(rho)	: 1.0000000000
Purity Tr(rho^2)	: 0.1488911426
Von Neumann entropy	: 1.9948940631
Eigenvalue min	: 9.344902e-02
Eigenvalue max	: 2.196529e-01
Imaginary part	: < 1e-6 (not shown)

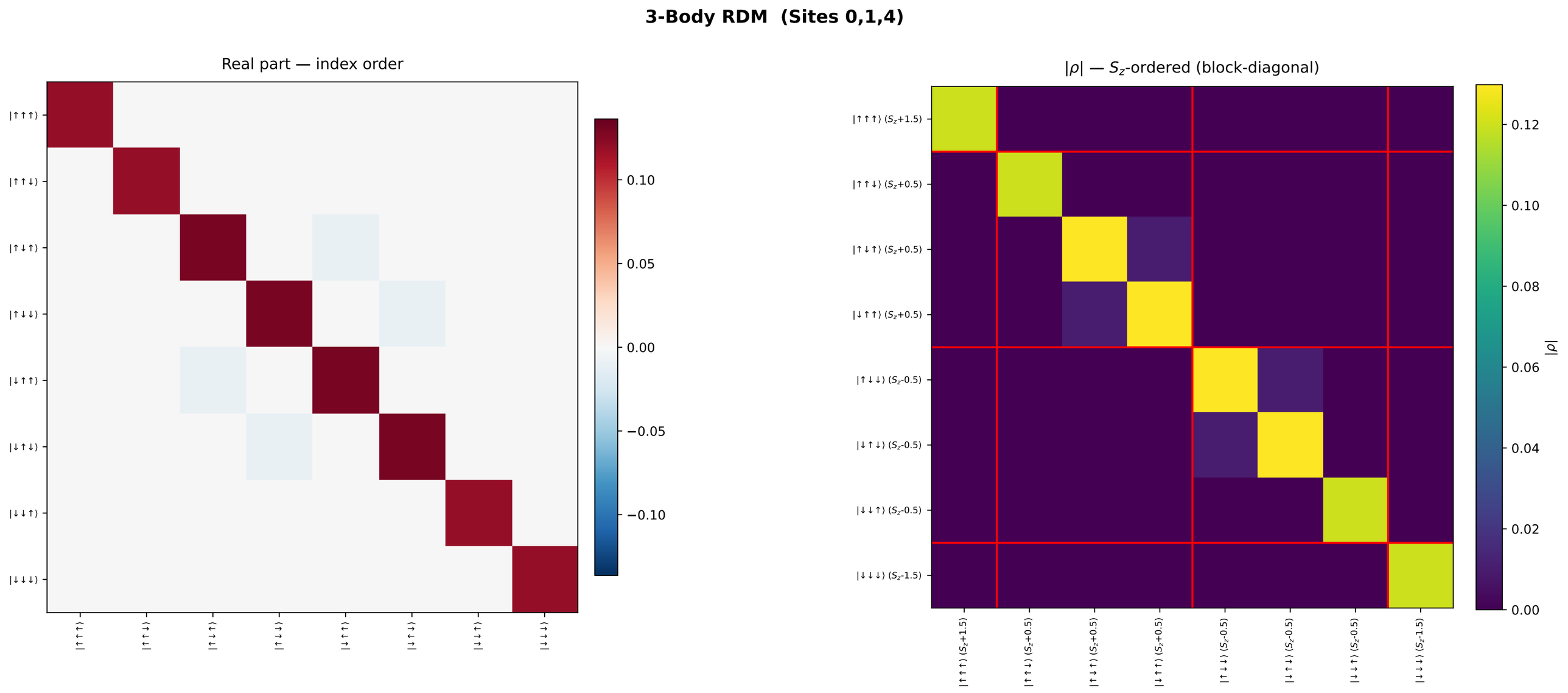
GME Analysis
==========

gamma = Tr(rho^2) = 0.14889114

A	B	Tr(rho_A^2)

STATUS: MIXED STATE

Tr(rho^2) < 1. Pure-state C_GME not applicable. See convex-roof result below.


3-Body RDM (Sites 0,1,4)
Real part — index order
|↑↑↑⟩
|↑↑↓⟩
|↑↓↑⟩
|↑↓↓⟩
|↓↑↑⟩
|↓↑↓⟩
|↓↓↑⟩
|↓↓↓⟩
0.10
0.05
0.00
−0.05
−0.10
|ρ| — S_z-ordered (block-diagonal)
|↑↑↑⟩ (S_z+1.5)
|↑↑↓⟩ (S_z+0.5)
|↑↓↑⟩ (S_z+0.5)
|↓↑↑⟩ (S_z+0.5)
|↑↓↓⟩ (S_z-0.5)
|↓↑↓⟩ (S_z-0.5)
|↓↓↑⟩ (S_z-0.5)
|↓↓↓⟩ (S_z-1.5)
0.12
0.10
0.08
0.06
0.04
0.02
0.00
|ρ|
Trace=1.000000 Purity=0.125559 Entropy=2.077259 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)

Analysis Summary - 3-Body RDM (Sites 0,1,4)

RDM Metrics
==========

Trace(rho)	: 1.0000000000
Purity Tr(rho^2)	: 0.1255590621
Von Neumann entropy	: 2.0772592054
Eigenvalue min	: 1.201735e-01
Eigenvalue max	: 1.394792e-01
Imaginary part	: < 1e-6 (not shown)

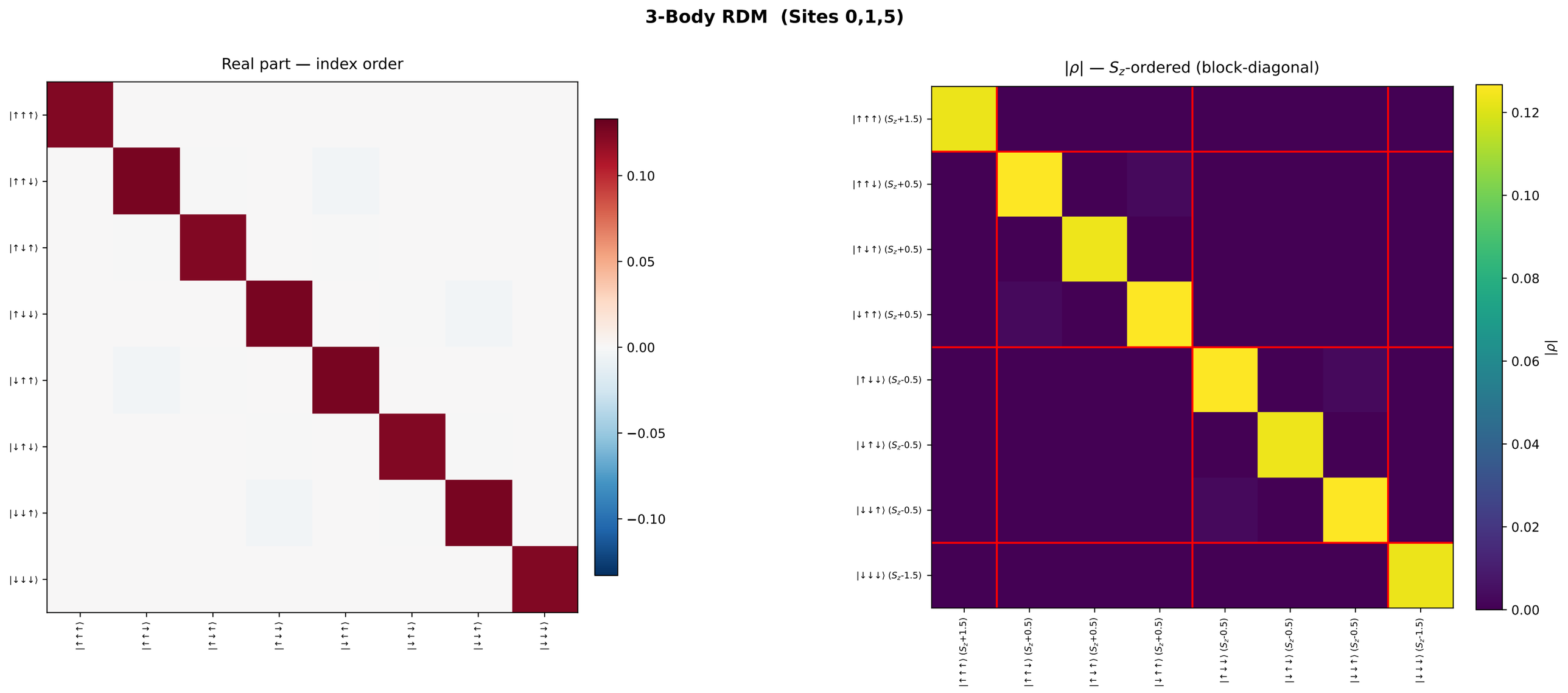
GME Analysis
==========

gamma = Tr(rho^2) = 0.12555906

A	B	Tr(rho_A^2)

STATUS: MIXED STATE

Tr(rho^2) < 1. Pure-state C_GME not applicable. See convex-roof result below.


3-Body RDM (Sites 0,1,5)
Real part — index order
|↑↑↑⟩
|↑↑↓⟩
|↑↓↑⟩
|↑↓↓⟩
|↓↑↑⟩
|↓↑↓⟩
|↓↓↑⟩
|↓↓↓⟩
0.10
0.05
0.00
−0.05
−0.10
|ρ| — S_z-ordered (block-diagonal)
|↑↑↑⟩ (S_z+1.5)
|↑↑↓⟩ (S_z+0.5)
|↑↓↑⟩ (S_z+0.5)
|↓↑↑⟩ (S_z+0.5)
|↑↓↓⟩ (S_z-0.5)
|↓↑↓⟩ (S_z-0.5)
|↓↓↑⟩ (S_z-0.5)
|↓↓↓⟩ (S_z-1.5)
0.12
0.10
0.08
0.06
0.04
0.02
0.00
|ρ|
Trace=1.000000 Purity=0.125067 Entropy=2.079175 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)

Analysis Summary - 3-Body RDM (Sites 0,1,5)

RDM Metrics
==========

Trace(rho)	: 1.0000000000
Purity Tr(rho^2)	: 0.1250673282
Von Neumann entropy	: 2.0791745511
Eigenvalue min	: 1.231782e-01
Eigenvalue max	: 1.300146e-01
Imaginary part	: < 1e-6 (not shown)

GME Analysis
==========

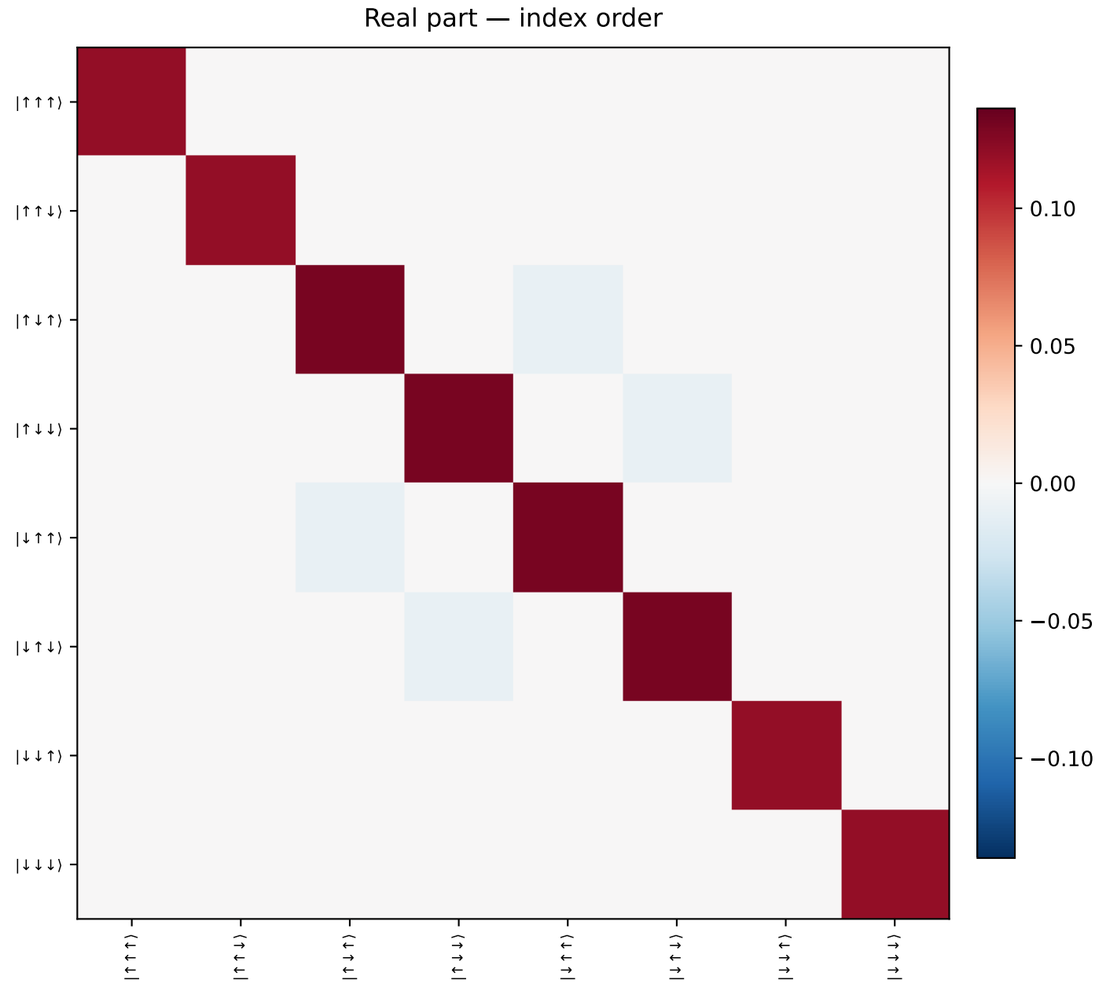
gamma = Tr(rho^2) = 0.12506733

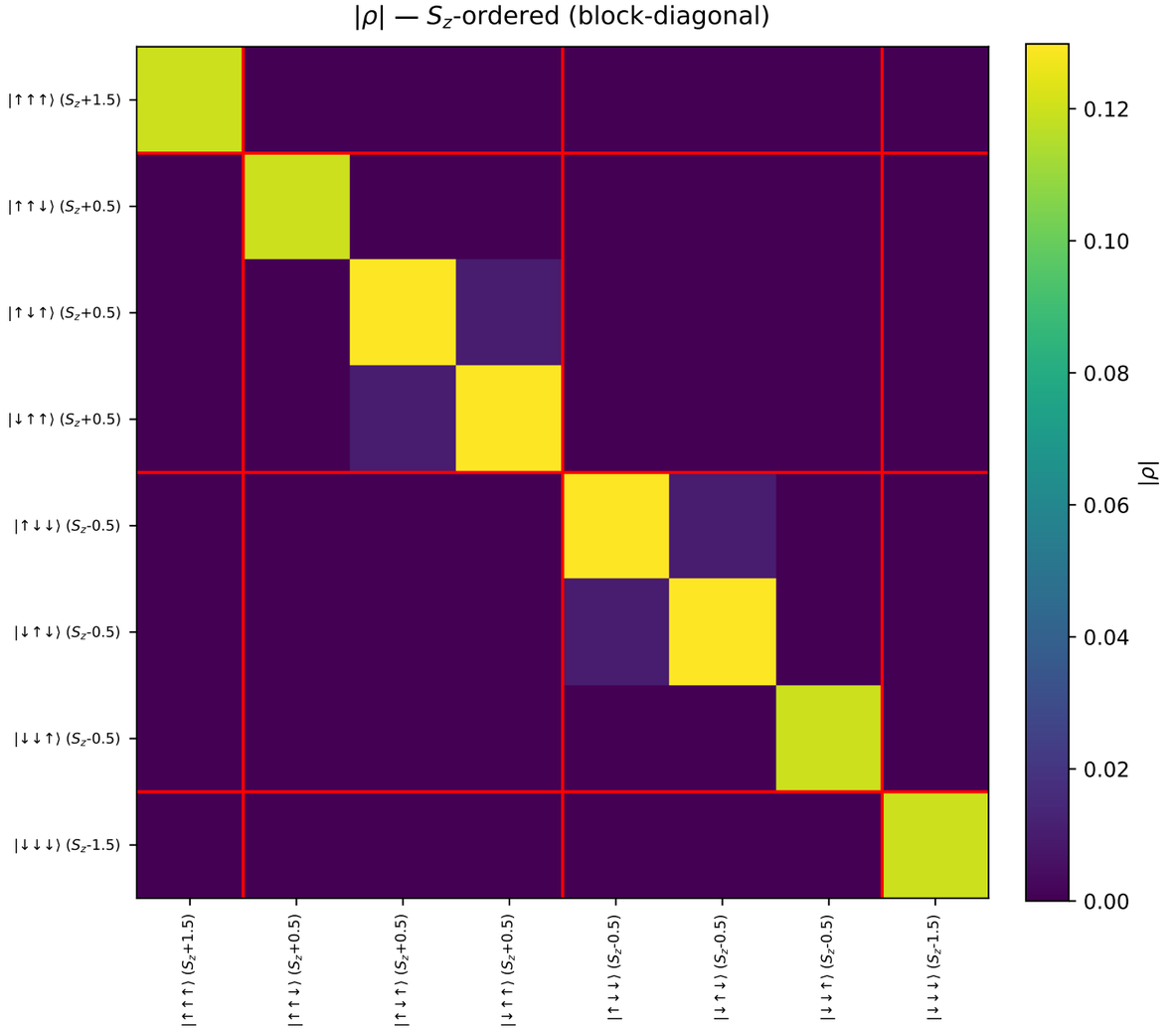
A	B	Tr(rho_A^2)

STATUS: MIXED STATE

Tr(rho^2) < 1. Pure-state C_GME not applicable. See convex-roof result below.

3-Body RDM (Sites 0,2,3)





Trace=1.000000 Purity=0.125559 Entropy=2.077261 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 ($\Rightarrow S_z$ conserved)

Analysis Summary - 3-Body RDM (Sites 0,2,3)

RDM Metrics
==========

Trace(rho)	: 1.0000000000
Purity Tr(rho^2)	: 0.1255585209
Von Neumann entropy	: 2.0772612935
Eigenvalue min	: 1.201759e-01
Eigenvalue max	: 1.394722e-01
Imaginary part	: < 1e-6 (not shown)

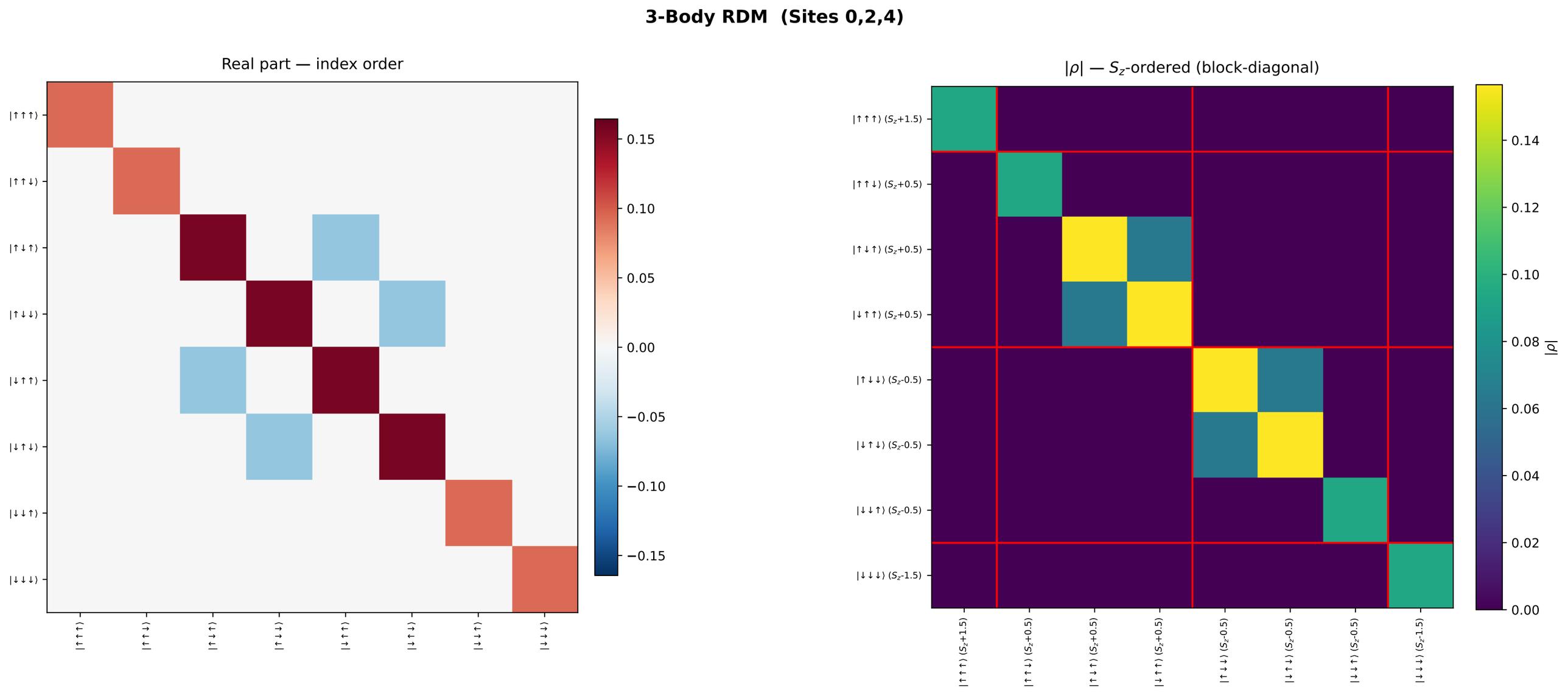
GME Analysis
==========

gamma = Tr(rho^2) = 0.12555852

A	B	Tr(rho_A^2)

STATUS: MIXED STATE

Tr(rho^2) < 1. Pure-state C_GME not applicable. See convex-roof result below.


3-Body RDM (Sites 0,2,4)
Real part — index order
|↑↑↑⟩
|↑↑↓⟩
|↑↓↑⟩
|↑↓↓⟩
|↓↑↑⟩
|↓↑↓⟩
|↓↓↑⟩
|↓↓↓⟩
0.15
0.10
0.05
0.00
−0.05
−0.10
−0.15
|ρ| — S_z-ordered (block-diagonal)
|↑↑↑⟩ (S_z+1.5)
|↑↑↓⟩ (S_z+0.5)
|↑↓↑⟩ (S_z+0.5)
|↓↑↑⟩ (S_z+0.5)
|↑↓↓⟩ (S_z-0.5)
|↓↑↓⟩ (S_z-0.5)
|↓↓↑⟩ (S_z-0.5)
|↓↓↓⟩ (S_z-1.5)
0.14
0.12
0.10
0.08
0.06
0.04
0.02
0.00
|ρ|
Trace=1.000000 Purity=0.148904 Entropy=1.994851 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)

Analysis Summary - 3-Body RDM (Sites 0,2,4)

RDM Metrics
==========

Trace(rho)	: 1.0000000000
Purity Tr(rho^2)	: 0.1489040060
Von Neumann entropy	: 1.9948505094
Eigenvalue min	: 9.344053e-02
Eigenvalue max	: 2.196784e-01
Imaginary part	: < 1e-6 (not shown)

GME Analysis
==========

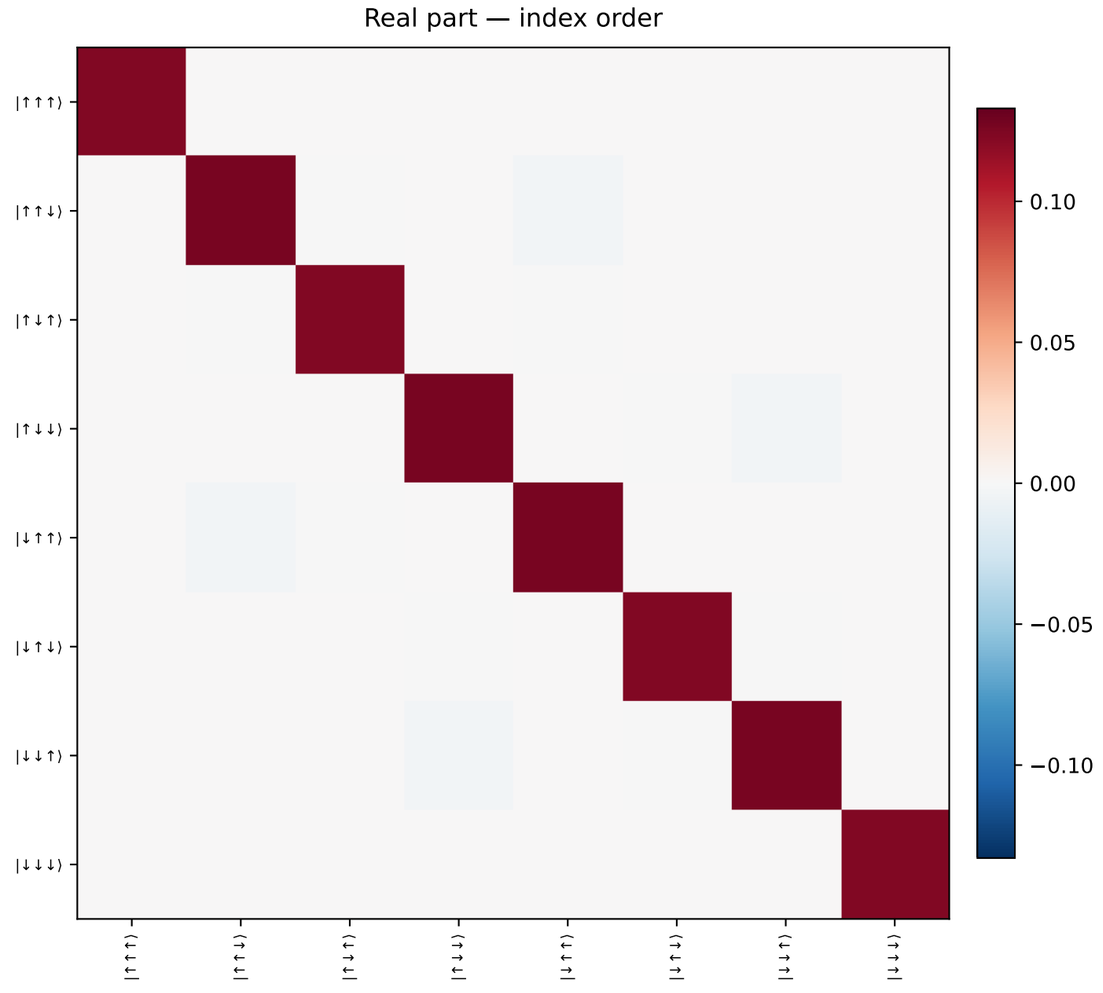
gamma = Tr(rho^2) = 0.14890401

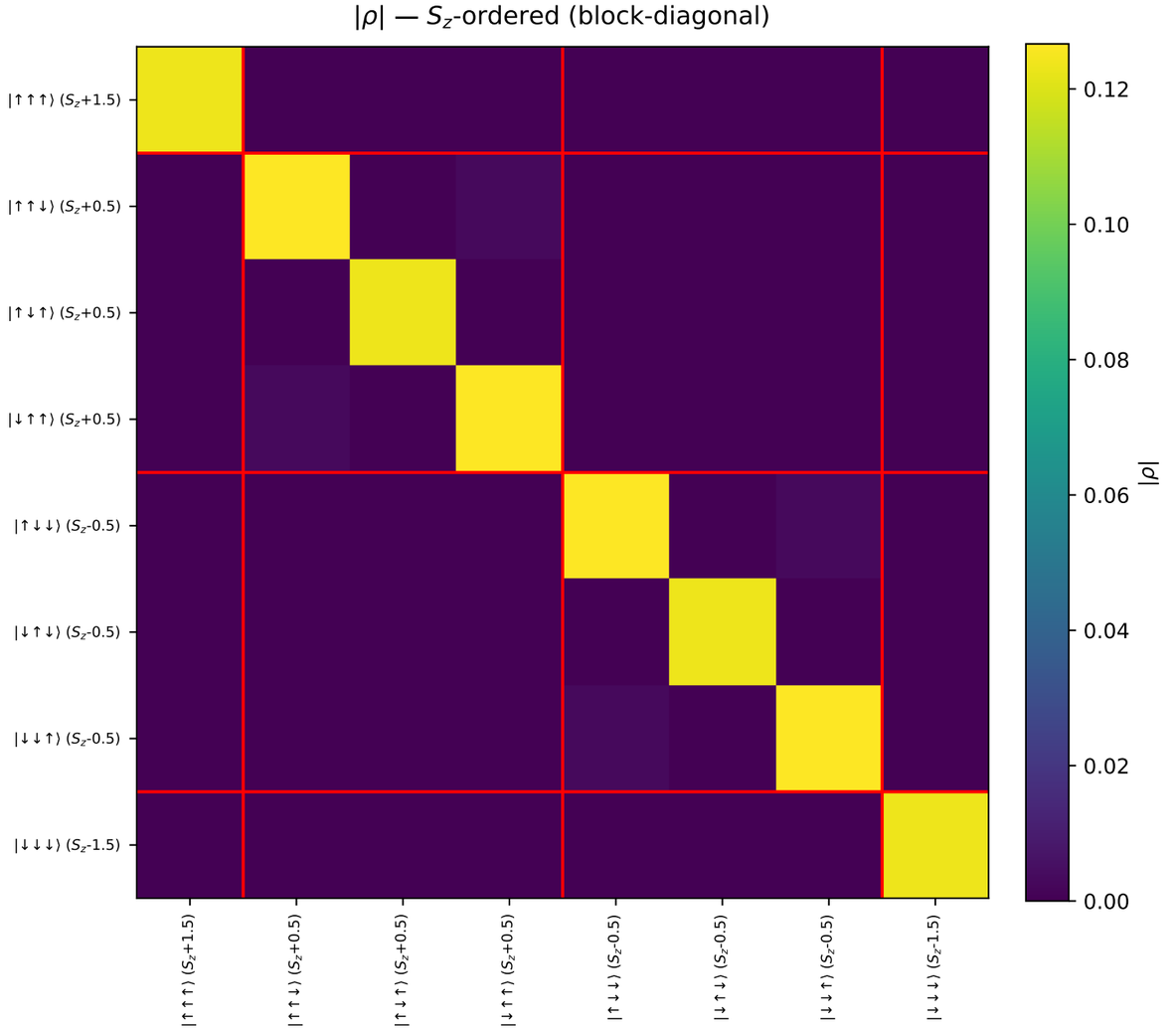
A	B	Tr(rho_A^2)

STATUS: MIXED STATE

Tr(rho^2) < 1. Pure-state C_GME not applicable. See convex-roof result below.

3-Body RDM (Sites 0,2,5)





Trace=1.000000 Purity=0.125065 Entropy=2.079184 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 ($\Rightarrow S_z$ conserved)

Analysis Summary - 3-Body RDM (Sites 0,2,5)

RDM Metrics
==========

Trace(rho)	: 1.0000000000
Purity Tr(rho^2)	: 0.1250649358
Von Neumann entropy	: 2.0791839988
Eigenvalue min	: 1.232101e-01
Eigenvalue max	: 1.299246e-01
Imaginary part	: < 1e-6 (not shown)

GME Analysis
==========

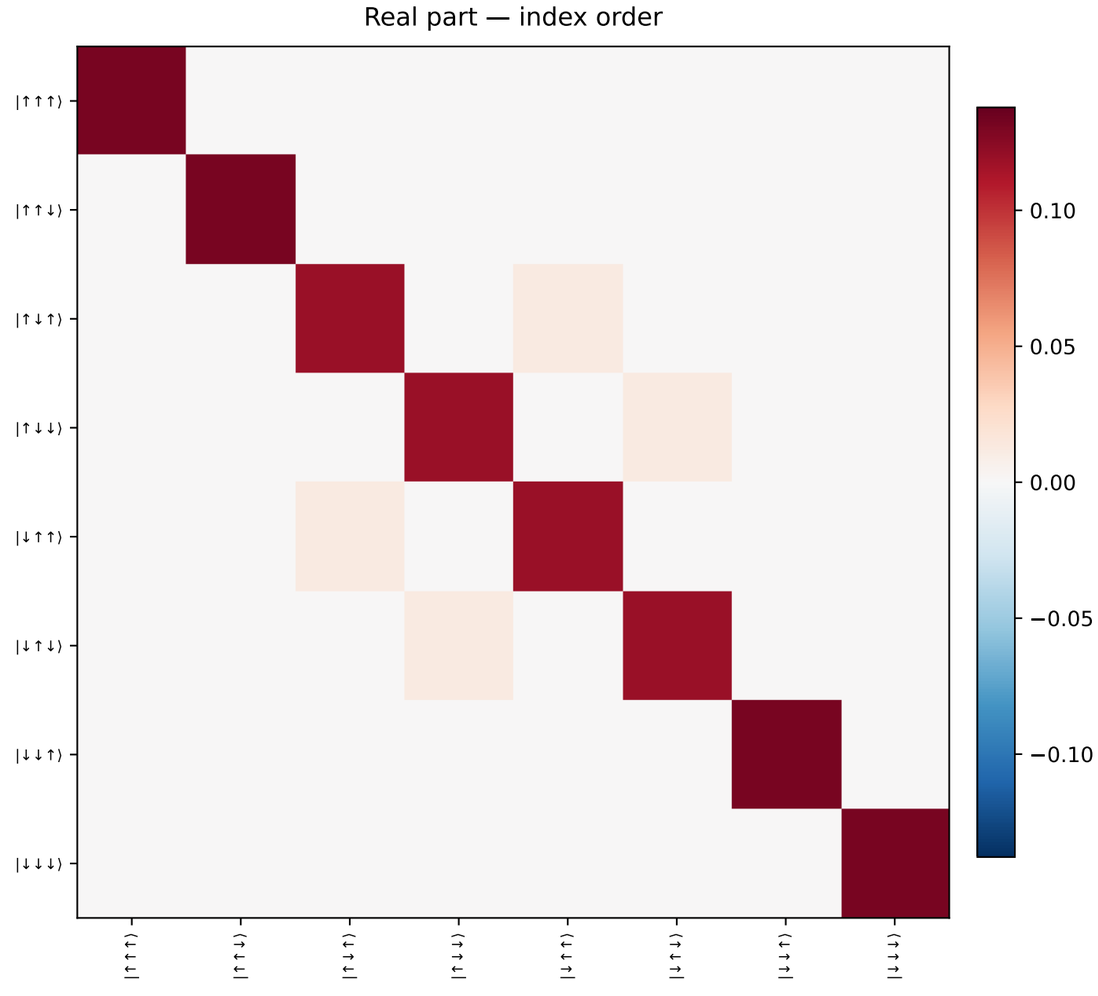
gamma = Tr(rho^2) = 0.12506494

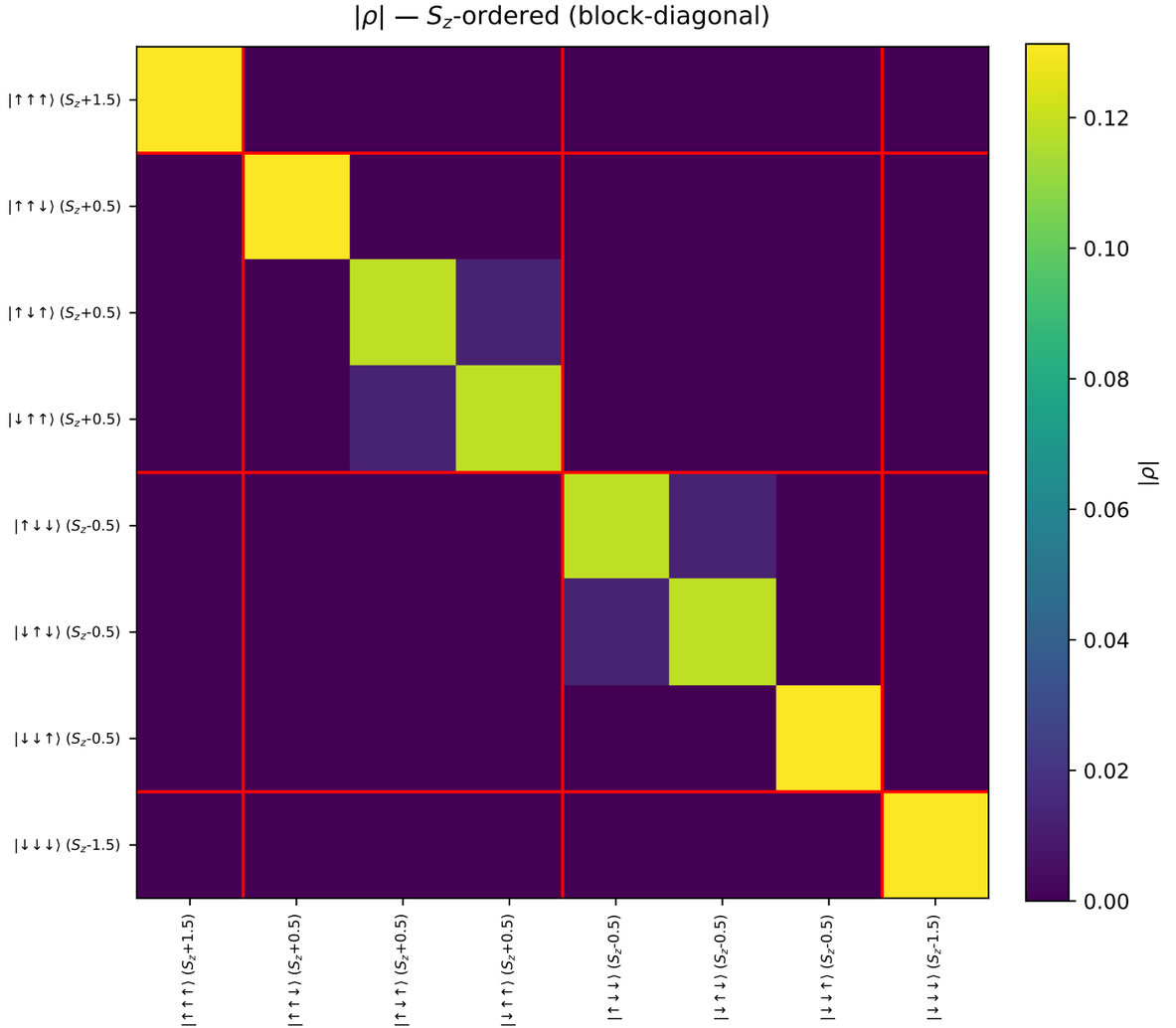
A	B	Tr(rho_A^2)

STATUS: MIXED STATE

Tr(rho^2) < 1. Pure-state C_GME not applicable. See convex-roof result below.

3-Body RDM (Sites 0,3,4)





Trace=1.000000 Purity=0.125949 Entropy=2.075505 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 ($\Rightarrow S_z$ conserved)

Analysis Summary - 3-Body RDM (Sites 0,3,4)

RDM Metrics
==========

Trace(rho)	: 1.0000000000
Purity Tr(rho^2)	: 0.1259492576
Von Neumann entropy	: 2.0755048607
Eigenvalue min	: 1.061328e-01
Eigenvalue max	: 1.312891e-01
Imaginary part	: < 1e-6 (not shown)

GME Analysis
==========

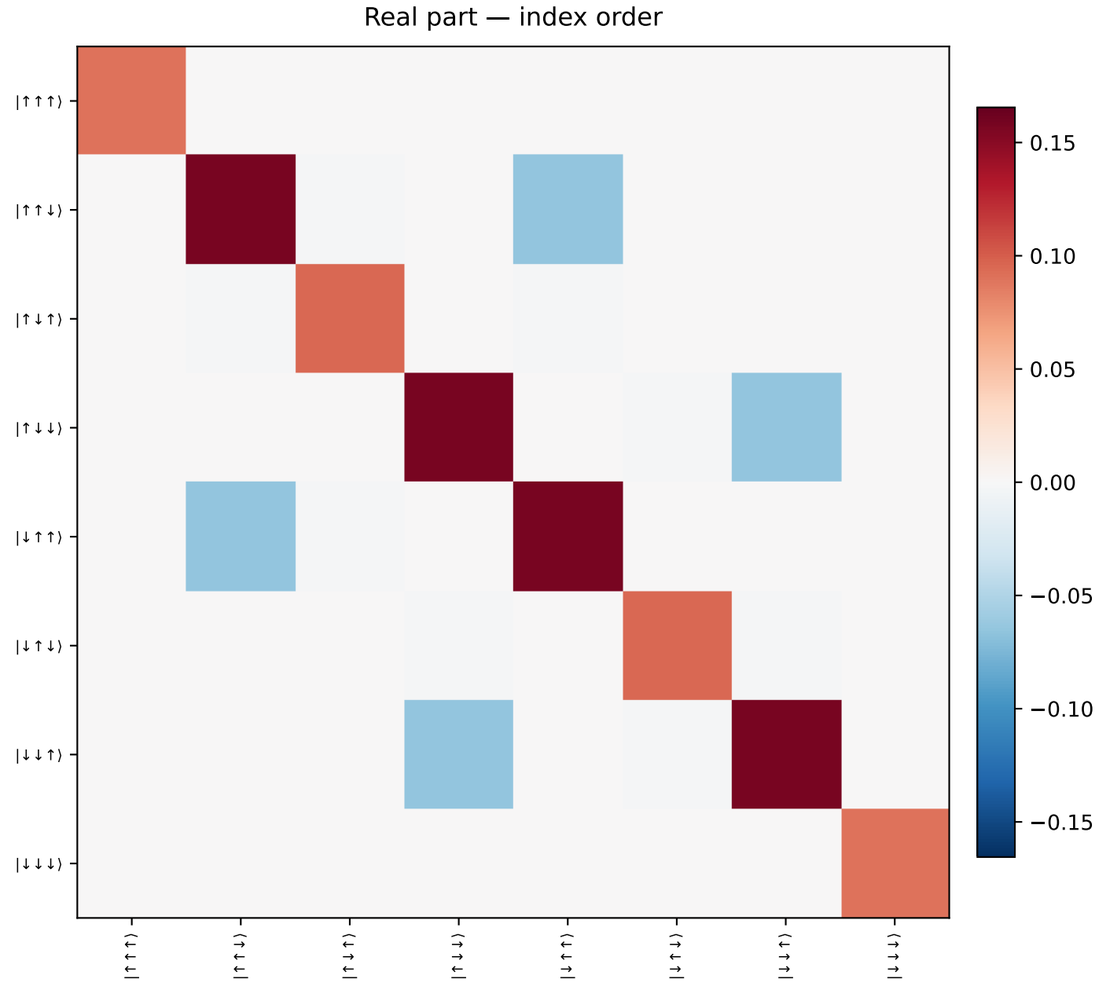
gamma = Tr(rho^2) = 0.12594926

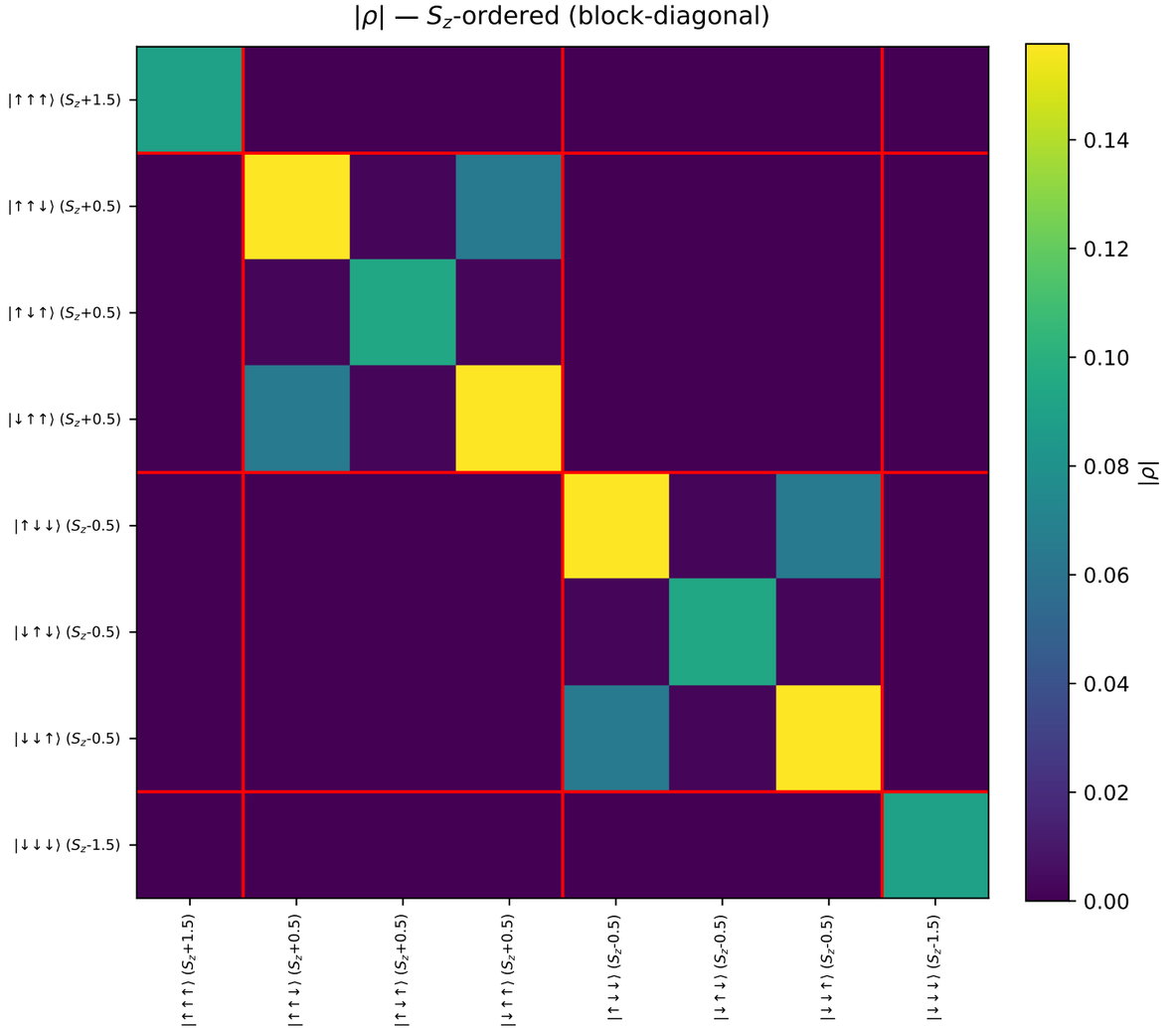
A	B	Tr(rho_A^2)

STATUS: MIXED STATE

Tr(rho^2) < 1. Pure-state C_GME not applicable. See convex-roof result below.

3-Body RDM (Sites 0,3,5)





Trace=1.000000 Purity=0.150430 Entropy=1.989597 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 ($\Rightarrow S_z$ conserved)

Analysis Summary - 3-Body RDM (Sites 0,3,5)

RDM Metrics
==========

Trace(rho)	: 1.0000000000
Purity Tr(rho^2)	: 0.1504300892
Von Neumann entropy	: 1.9895972634
Eigenvalue min	: 9.049354e-02
Eigenvalue max	: 2.225630e-01
Imaginary part	: < 1e-6 (not shown)

GME Analysis
==========

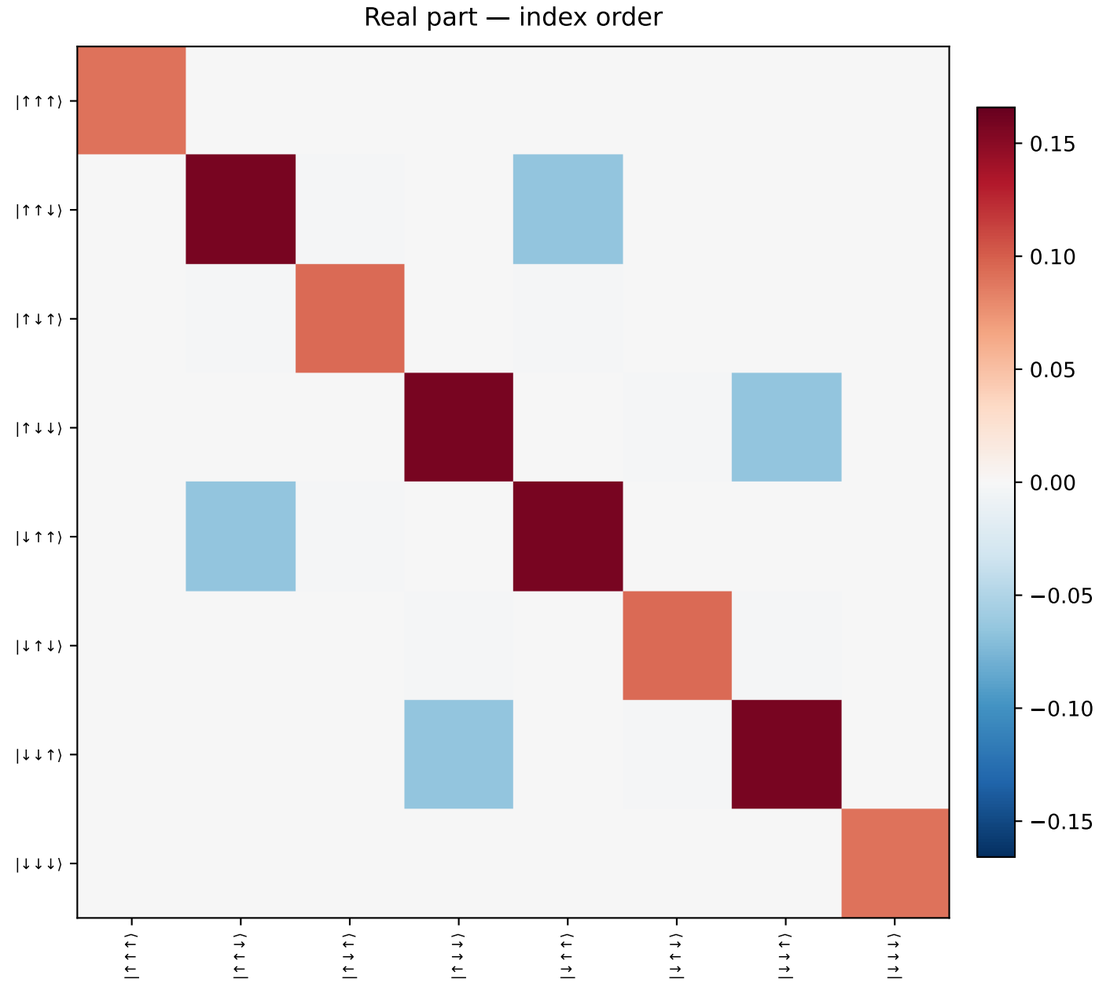
gamma = Tr(rho^2) = 0.15043009

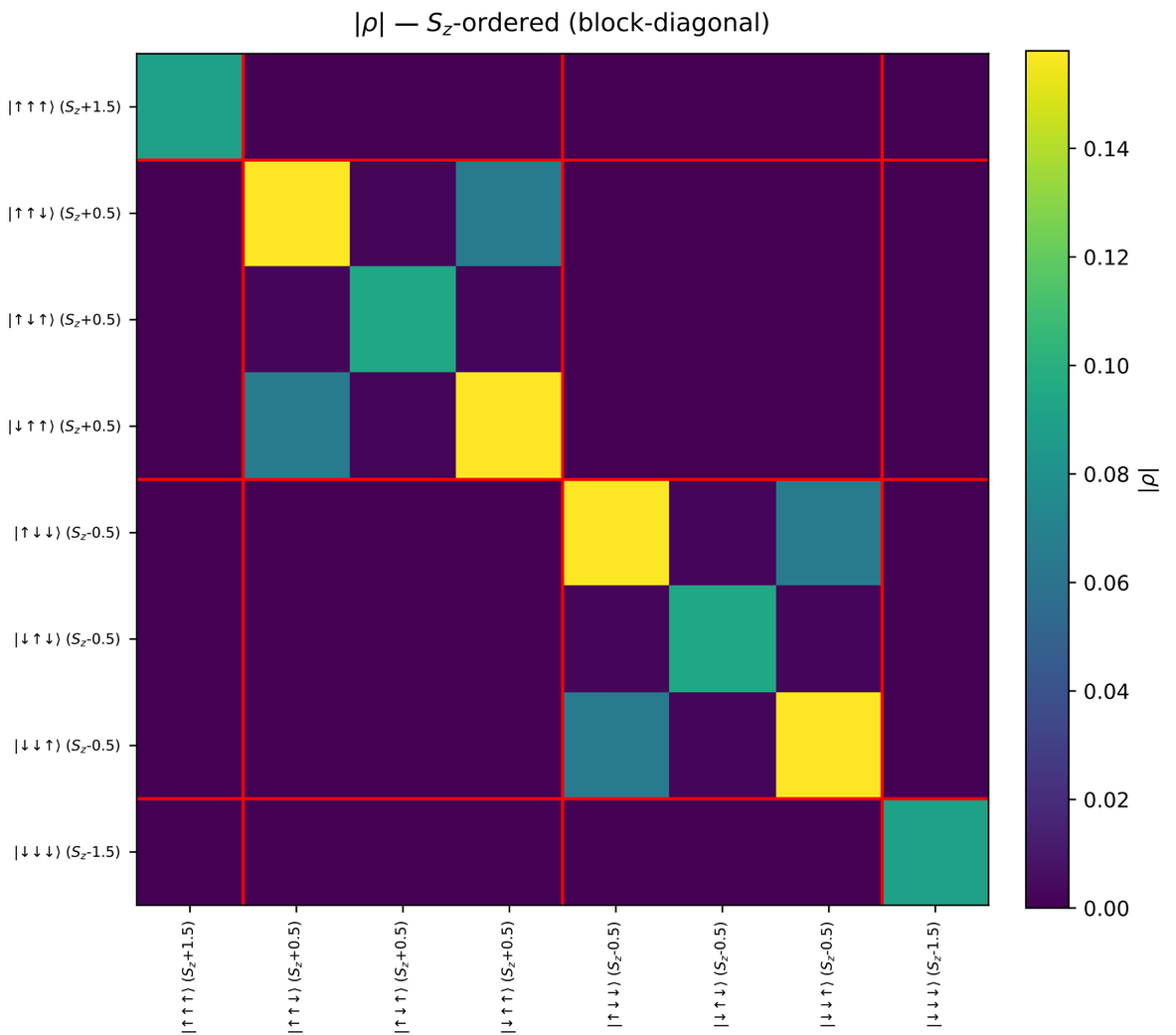
A	B	Tr(rho_A^2)

STATUS: MIXED STATE

Tr(rho^2) < 1. Pure-state C_GME not applicable. See convex-roof result below.

3-Body RDM (Sites 0,4,5)





Trace=1.000000 Purity=0.150975 Entropy=1.987757 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 ($\Rightarrow S_z$ conserved)

Analysis Summary - 3-Body RDM (Sites 0,4,5)

RDM Metrics
==========

Trace(rho)	: 1.0000000000
Purity Tr(rho^2)	: 0.1509750785
Von Neumann entropy	: 1.9877574325
Eigenvalue min	: 9.014436e-02
Eigenvalue max	: 2.236047e-01
Imaginary part	: < 1e-6 (not shown)

GME Analysis
==========

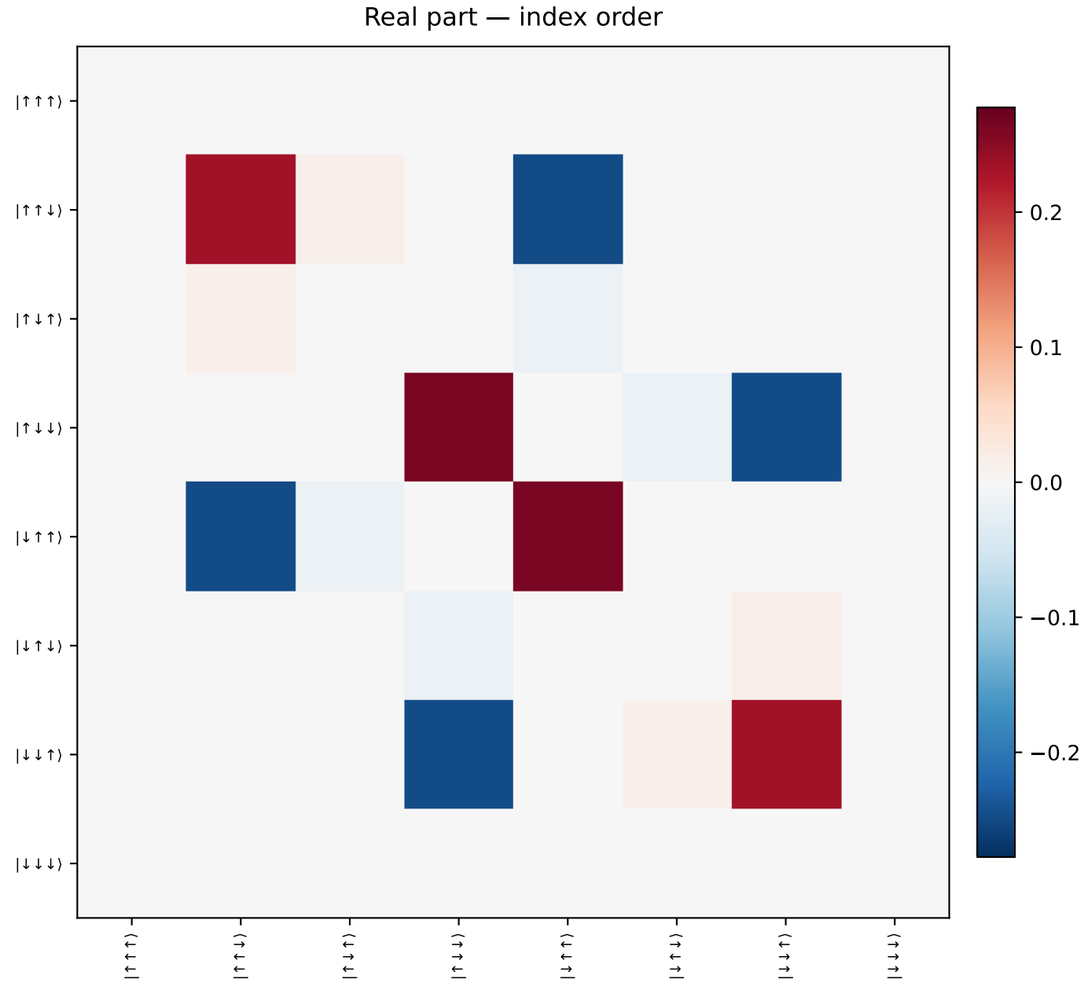
gamma = Tr(rho^2) = 0.15097508

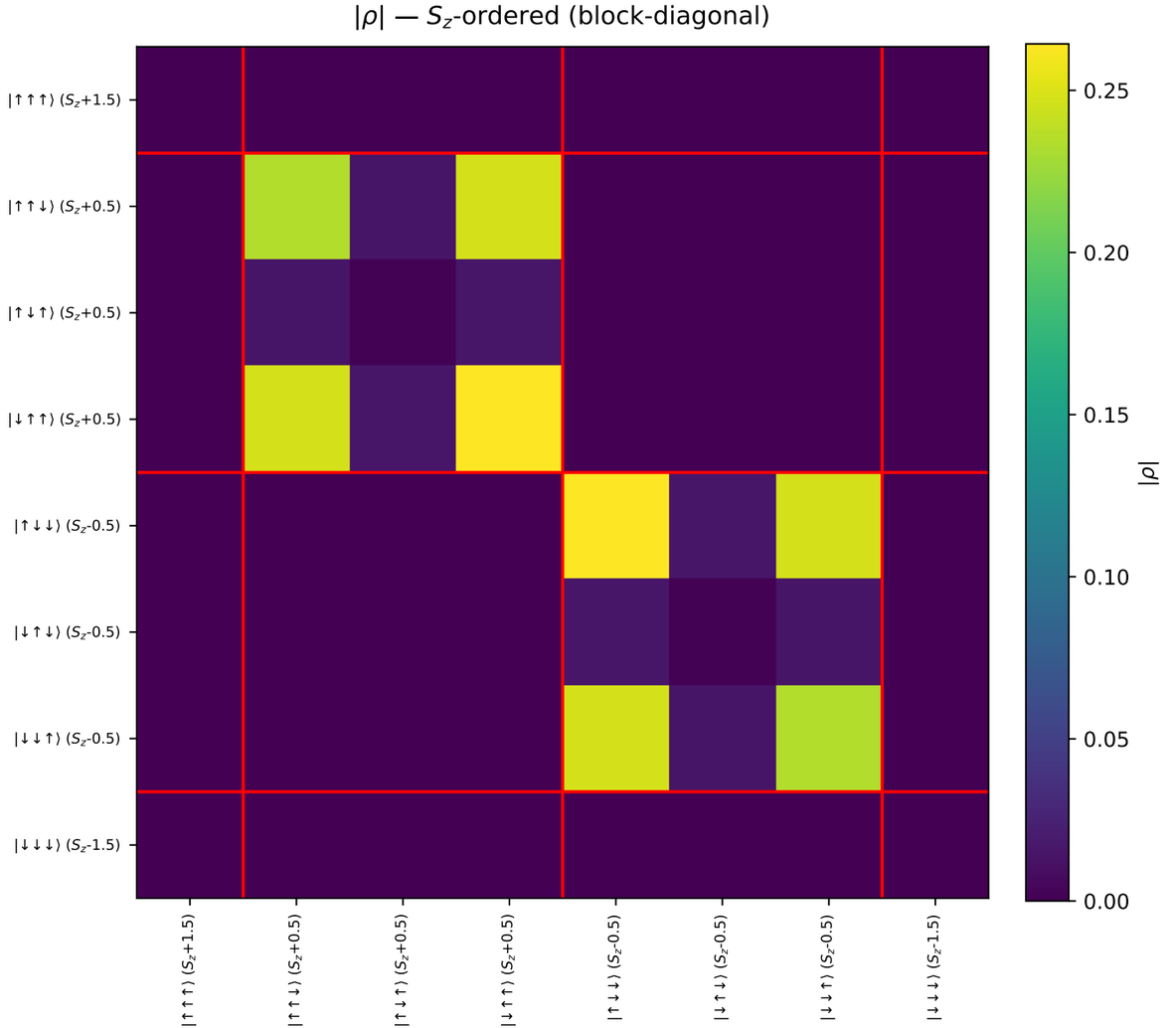
A	B	Tr(rho_A^2)

STATUS: MIXED STATE

Tr(rho^2) < 1. Pure-state C_GME not applicable. See convex-roof result below.

3-Body RDM (Sites 1,2,3)





Trace=1.000000 Purity=0.496207 Entropy=0.722312 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 ($\Rightarrow S_z$ conserved)

Analysis Summary - 3-Body RDM (Sites 1,2,3)

RDM Metrics
==========

Trace(rho)	: 1.0000000000
Purity Tr(rho^2)	: 0.4962065490
Von Neumann entropy	: 0.7223117676
Eigenvalue min	: 6.338489e-04
Eigenvalue max	: 4.980985e-01
Imaginary part	: < 1e-6 (not shown)

GME Analysis
==========

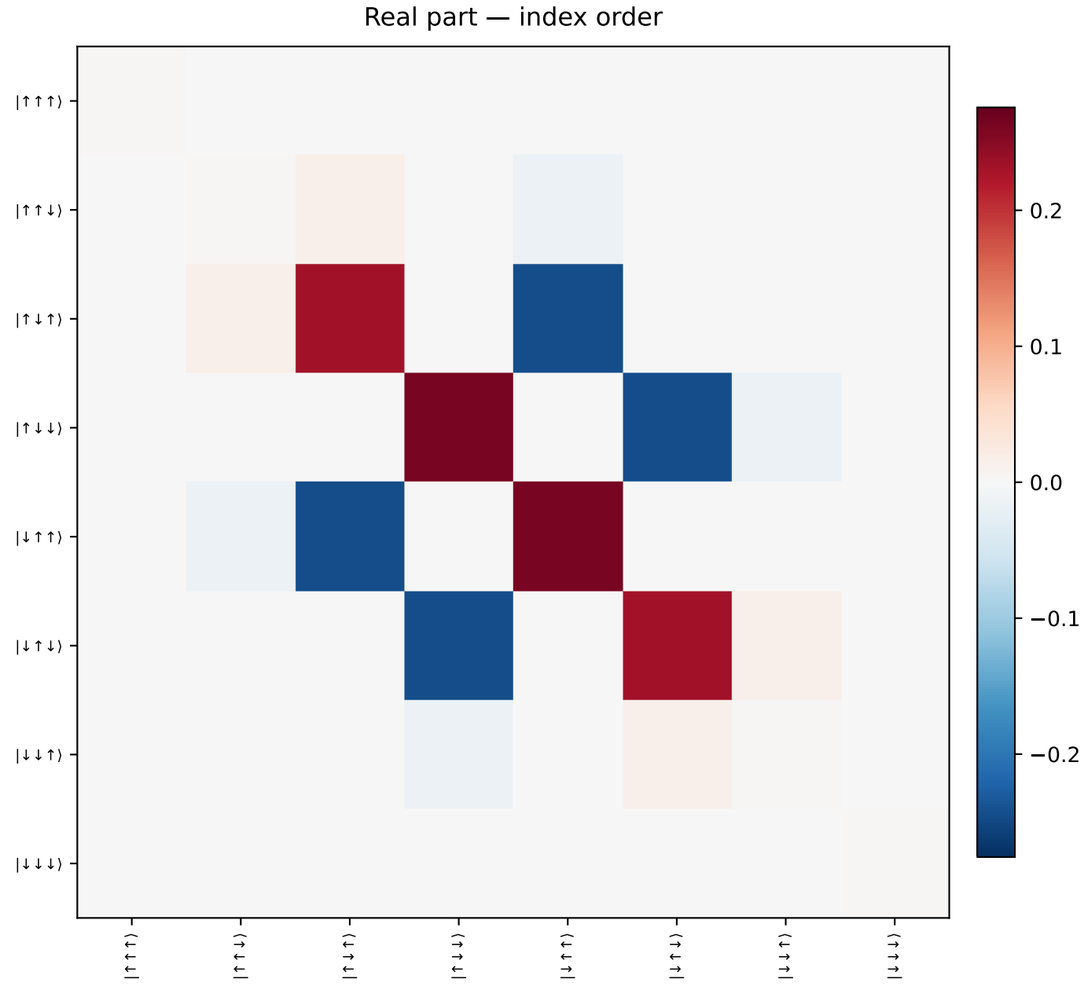
gamma = Tr(rho^2) = 0.49620655

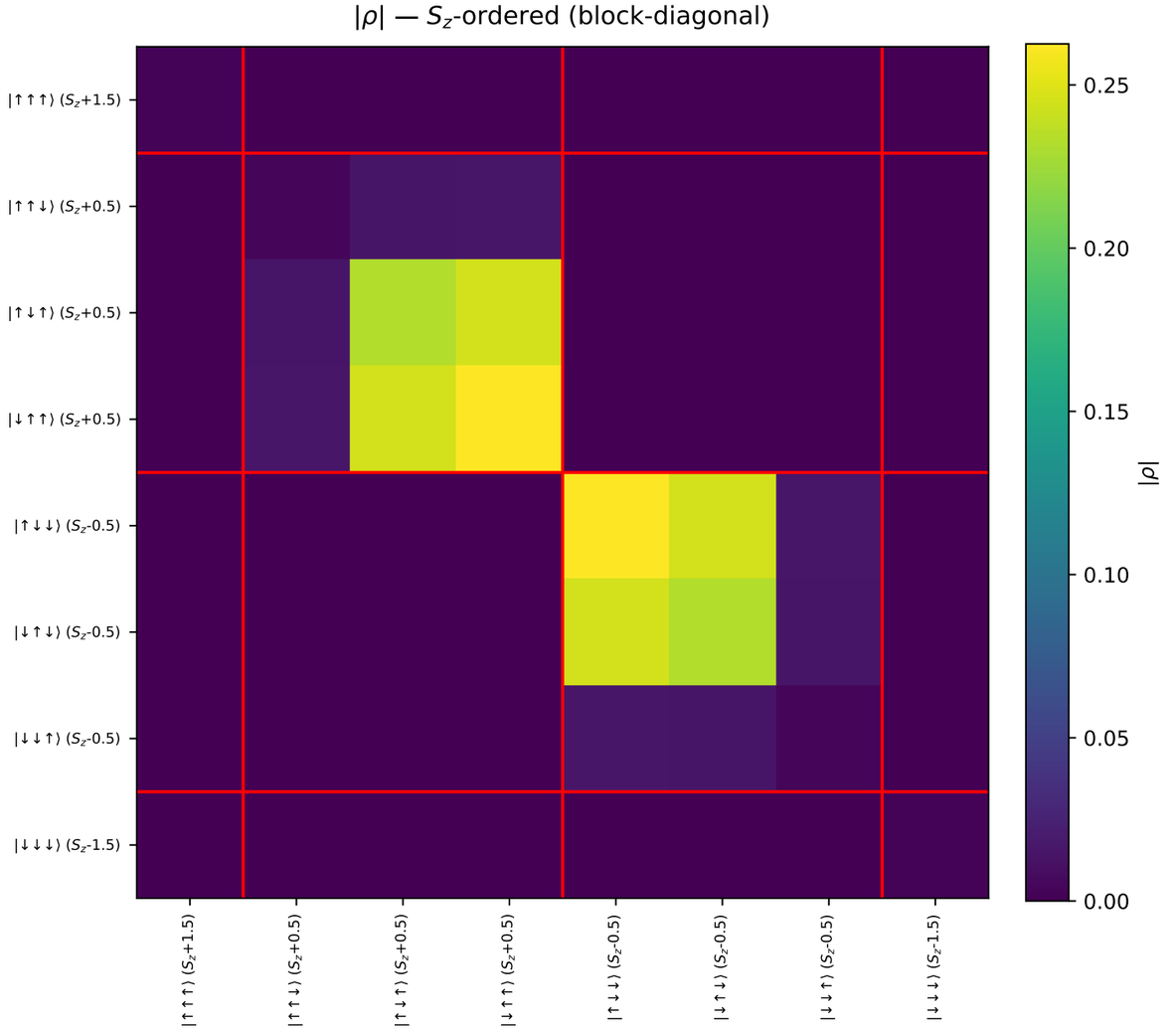
A	B	Tr(rho_A^2)

STATUS: MIXED STATE

Tr(rho^2) < 1. Pure-state C_GME not applicable. See convex-roof result below.

3-Body RDM (Sites 1,2,4)





Trace=1.000000 Purity=0.486887 Entropy=0.778046 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 ($\Rightarrow S_z$ conserved)

Analysis Summary - 3-Body RDM (Sites 1,2,4)

RDM Metrics
==========

Trace(rho)	: 1.0000000000
Purity Tr(rho^2)	: 0.4868869169
Von Neumann entropy	: 0.7780459582
Eigenvalue min	: 2.204961e-03
Eigenvalue max	: 4.933851e-01
Imaginary part	: < 1e-6 (not shown)

GME Analysis
==========

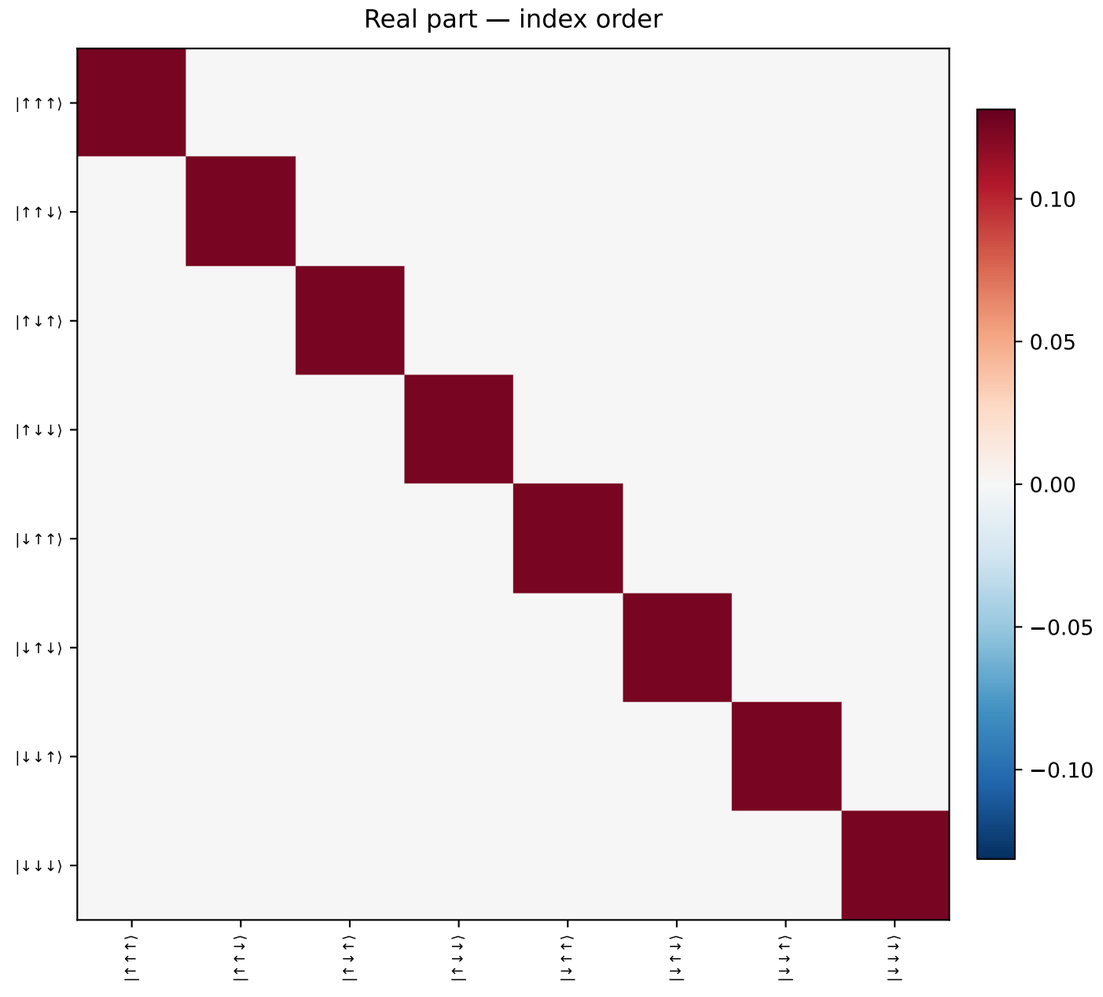
gamma = Tr(rho^2) = 0.48688692

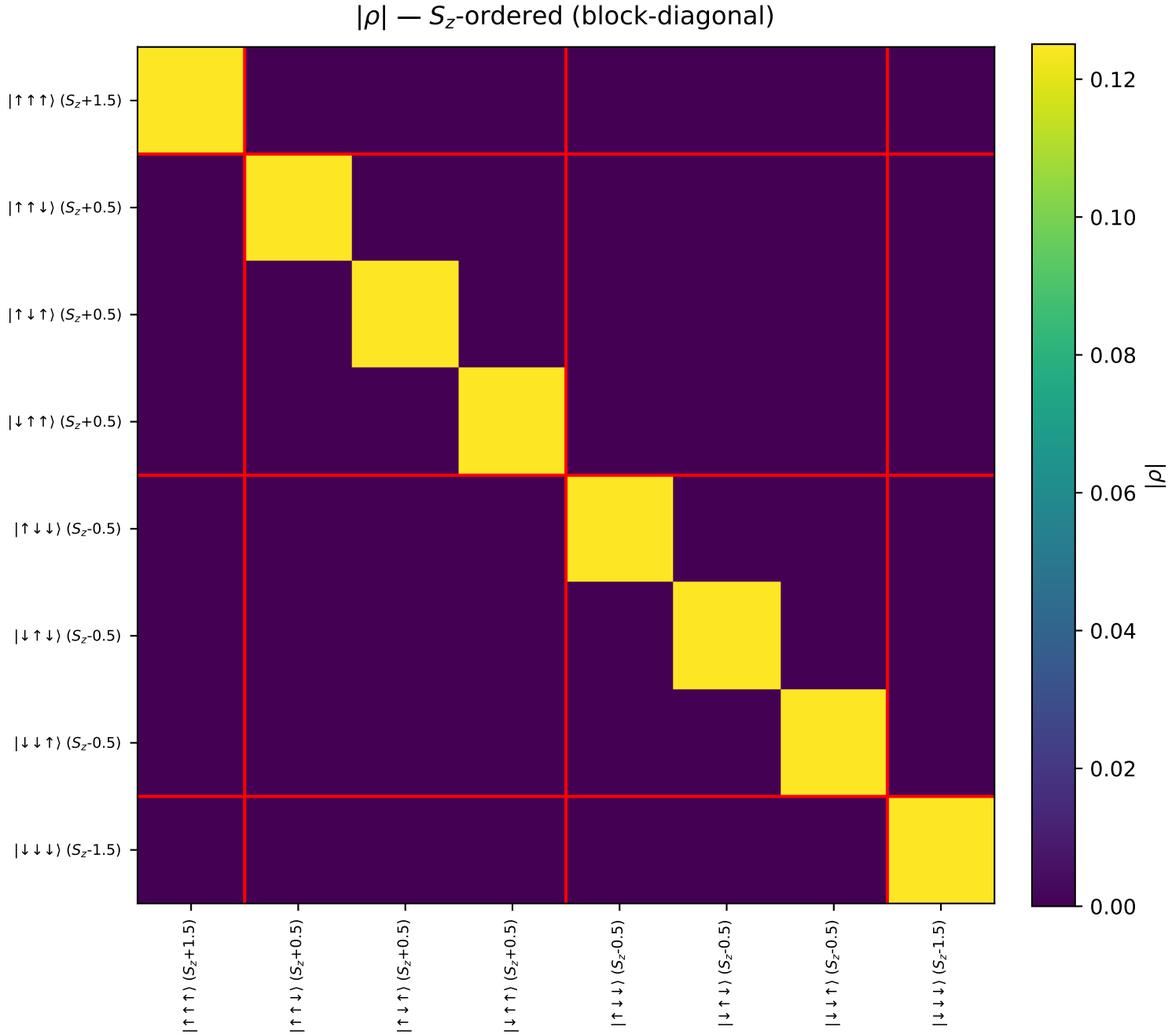
A	B	Tr(rho_A^2)

STATUS: MIXED STATE

Tr(rho^2) < 1. Pure-state C_GME not applicable. See convex-roof result below.

3-Body RDM (Sites 1,2,5)





Trace=1.000000 Purity=0.125000 Entropy=2.079441 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 ($\Rightarrow S_z$ conserved)

Analysis Summary - 3-Body RDM (Sites 1,2,5)

RDM Metrics
==========

Trace(rho)	: 1.0000000000
Purity Tr(rho^2)	: 0.1250001535
Von Neumann entropy	: 2.0794409280
Eigenvalue min	: 1.249047e-01
Eigenvalue max	: 1.252374e-01
Imaginary part	: < 1e-6 (not shown)

GME Analysis
==========

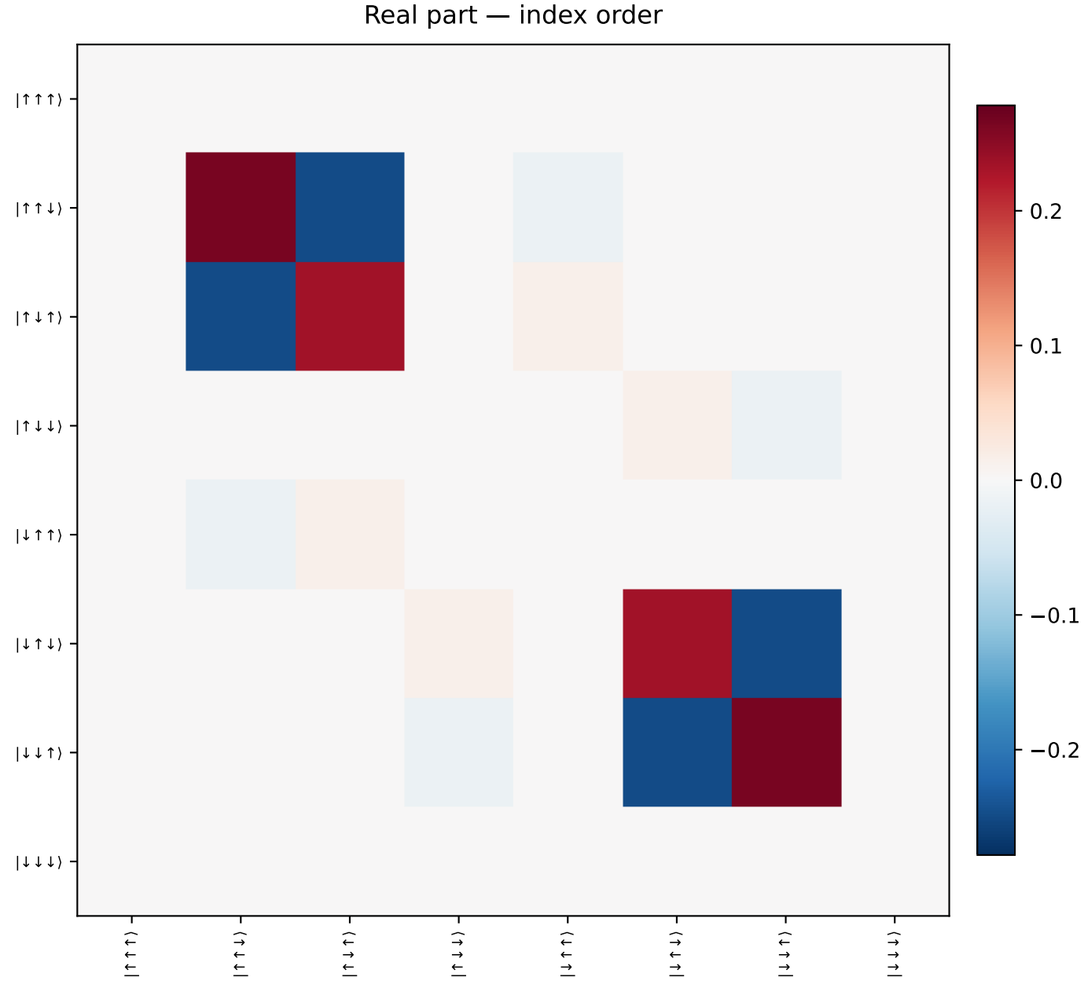
gamma = Tr(rho^2) = 0.12500015

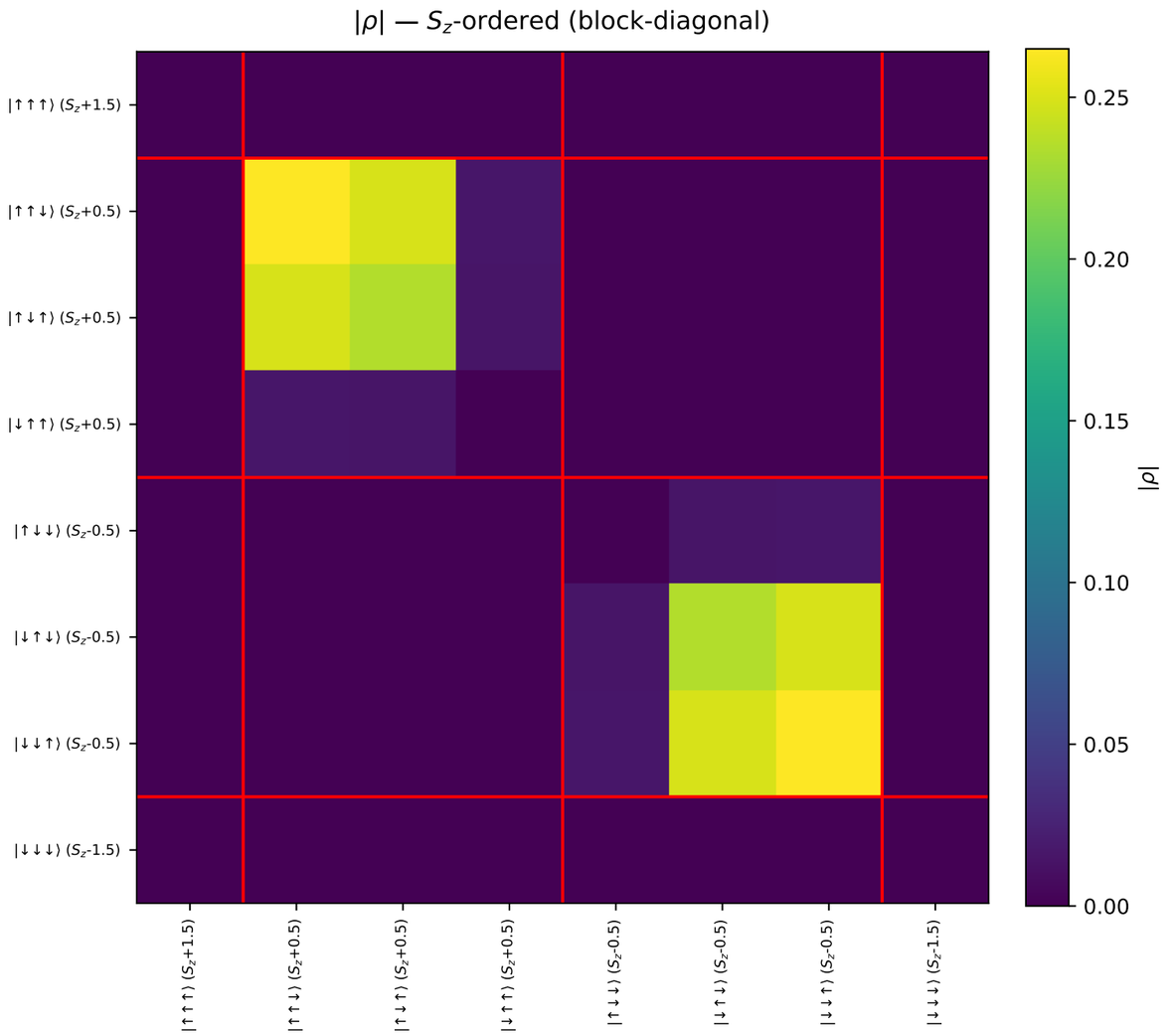
A	B	Tr(rho_A^2)

STATUS: MIXED STATE

Tr(rho^2) < 1. Pure-state C_GME not applicable. See convex-roof result below.

3-Body RDM (Sites 1,3,4)





Trace=1.000000 Purity=0.499664 Entropy=0.696539 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 ($\Rightarrow S_z$ conserved)

Analysis Summary - 3-Body RDM (Sites 1,3,4)

RDM Metrics
==========

Trace(rho)	: 1.0000000000
Purity Tr(rho^2)	: 0.4996641682
Von Neumann entropy	: 0.6965387921
Eigenvalue min	: 5.510387e-05
Eigenvalue max	: 4.998320e-01
Imaginary part	: < 1e-6 (not shown)

GME Analysis
==========

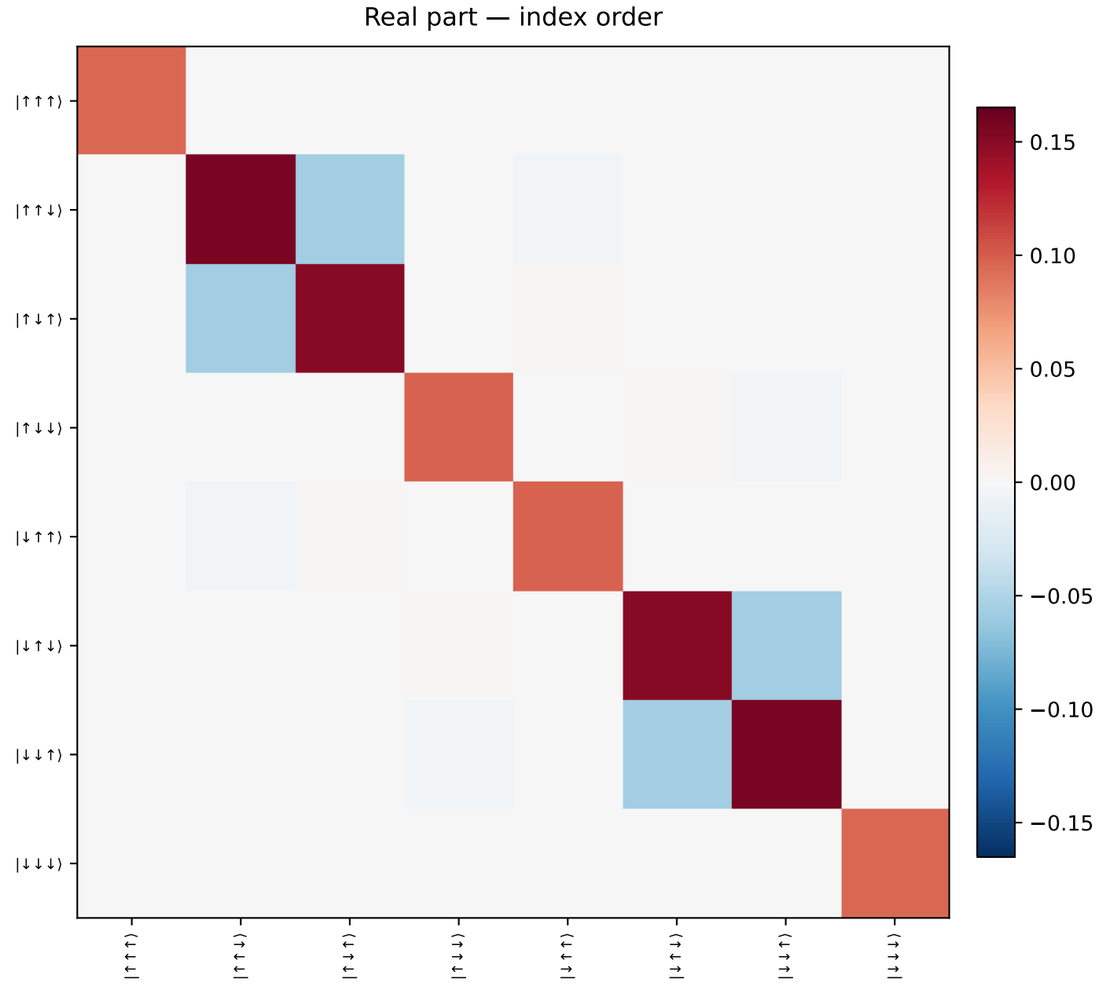
gamma = Tr(rho^2) = 0.49966417

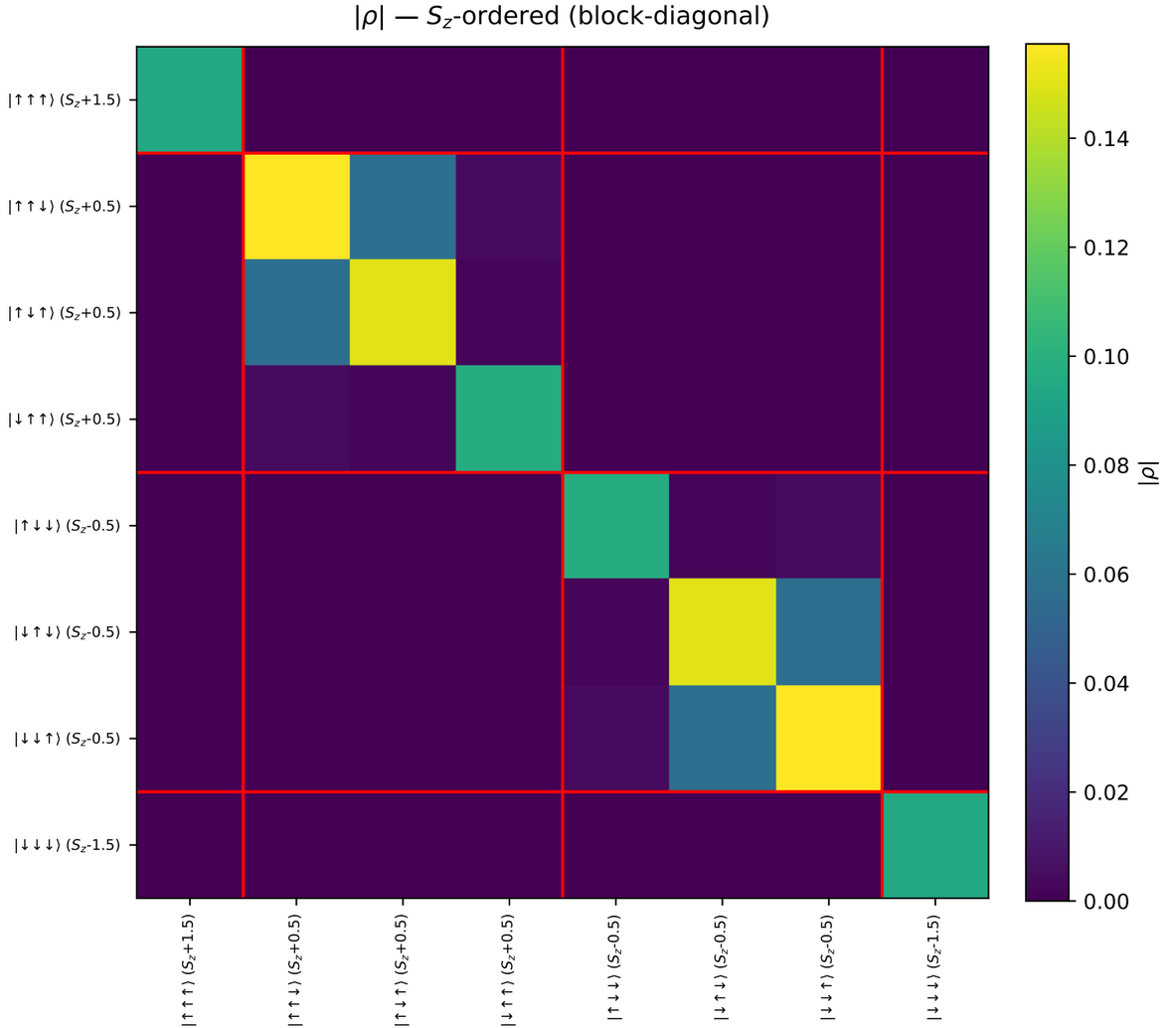
A	B	Tr(rho_A^2)

STATUS: MIXED STATE

Tr(rho^2) < 1. Pure-state C_GME not applicable. See convex-roof result below.

3-Body RDM (Sites 1,3,5)





Trace=1.000000 Purity=0.145172 Entropy=2.007503 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 ($\Rightarrow S_z$ conserved)

Analysis Summary - 3-Body RDM (Sites 1,3,5)

RDM Metrics
==========

Trace(rho)	: 1.0000000000
Purity Tr(rho^2)	: 0.1451716469
Von Neumann entropy	: 2.0075033507
Eigenvalue min	: 9.470402e-02
Eigenvalue max	: 2.119283e-01
Imaginary part	: < 1e-6 (not shown)

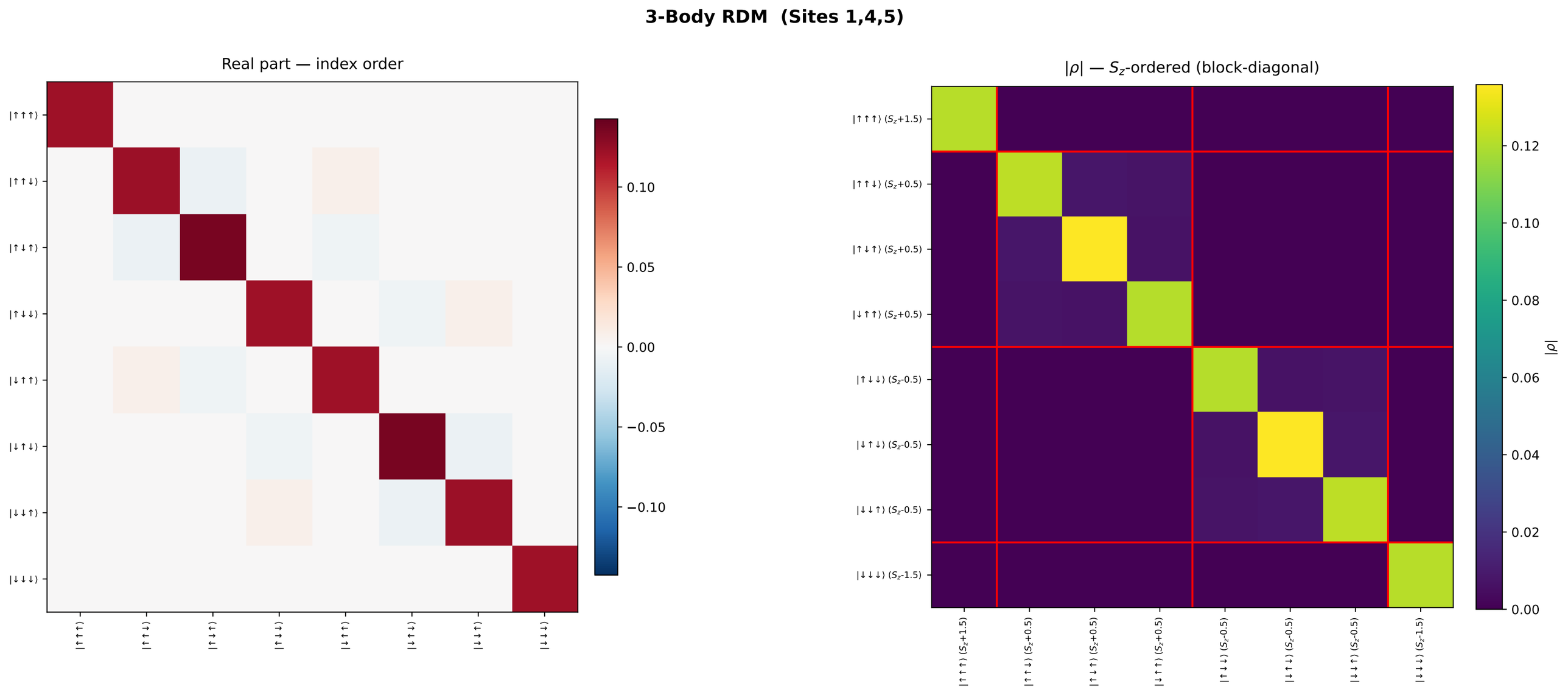
GME Analysis
==========

gamma = Tr(rho^2) = 0.14517165

A	B	Tr(rho_A^2)

STATUS: MIXED STATE

Tr(rho^2) < 1. Pure-state C_GME not applicable. See convex-roof result below.


3-Body RDM (Sites 1,4,5)
Real part — index order
|↑↑↑⟩
|↑↑↓⟩
|↑↓↑⟩
|↑↓↓⟩
|↓↑↑⟩
|↓↑↓⟩
|↓↓↑⟩
|↓↓↓⟩
0.10
0.05
0.00
−0.05
−0.10
|ρ| — S_z-ordered (block-diagonal)
|↑↑↑⟩ (S_z+1.5)
|↑↑↓⟩ (S_z+0.5)
|↑↓↑⟩ (S_z+0.5)
|↓↑↑⟩ (S_z+0.5)
|↑↓↓⟩ (S_z-0.5)
|↓↑↓⟩ (S_z-0.5)
|↓↓↑⟩ (S_z-0.5)
|↓↓↓⟩ (S_z-1.5)
0.12
0.10
0.08
0.06
0.04
0.02
0.00
|ρ|
Trace=1.000000 Purity=0.125941 Entropy=2.075762 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)

Analysis Summary - 3-Body RDM (Sites 1,4,5)

RDM Metrics
==========

Trace(rho)	: 1.0000000000
Purity Tr(rho^2)	: 0.1259414218
Von Neumann entropy	: 2.0757624647
Eigenvalue min	: 1.141110e-01
Eigenvalue max	: 1.430661e-01
Imaginary part	: < 1e-6 (not shown)

GME Analysis
==========

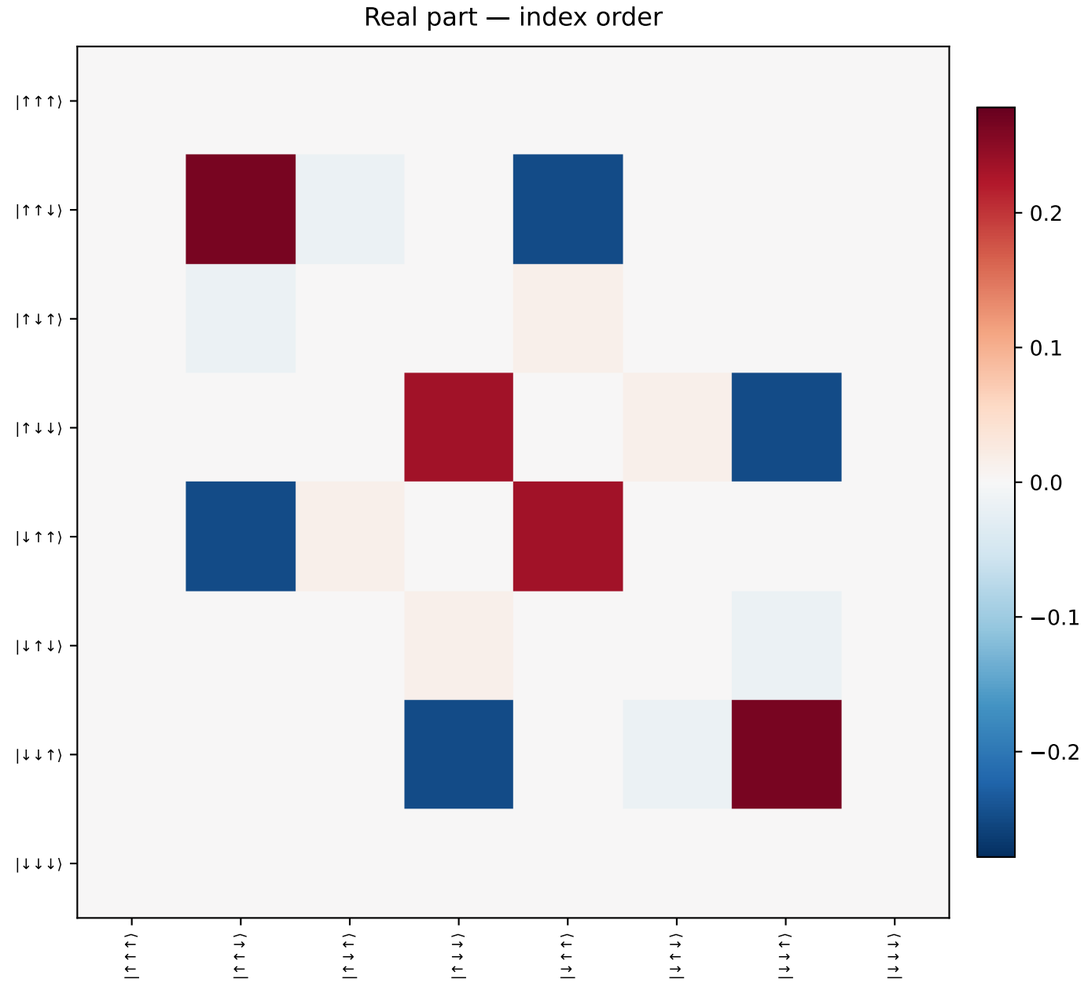
gamma = Tr(rho^2) = 0.12594142

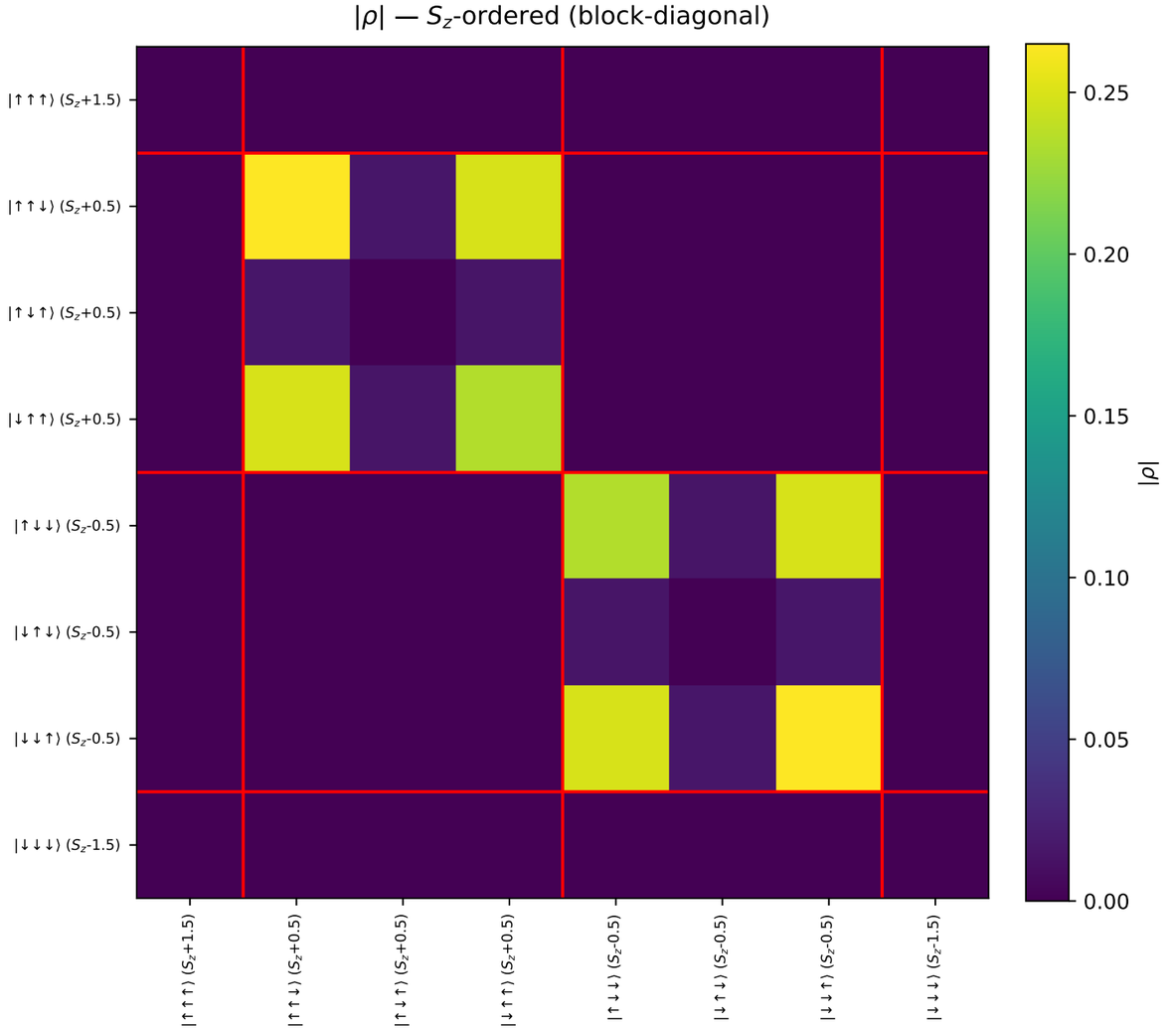
A	B	Tr(rho_A^2)

STATUS: MIXED STATE

Tr(rho^2) < 1. Pure-state C_GME not applicable. See convex-roof result below.

3-Body RDM (Sites 2,3,4)





Trace=1.000000 Purity=0.499782 Entropy=0.695444 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 ($\Rightarrow S_z$ conserved)

Analysis Summary - 3-Body RDM (Sites 2,3,4)

RDM Metrics
==========

Trace(rho)	: 1.0000000000
Purity Tr(rho^2)	: 0.4997819190
Von Neumann entropy	: 0.6954436893
Eigenvalue min	: 3.635211e-05
Eigenvalue max	: 4.998909e-01
Imaginary part	: < 1e-6 (not shown)

GME Analysis
==========

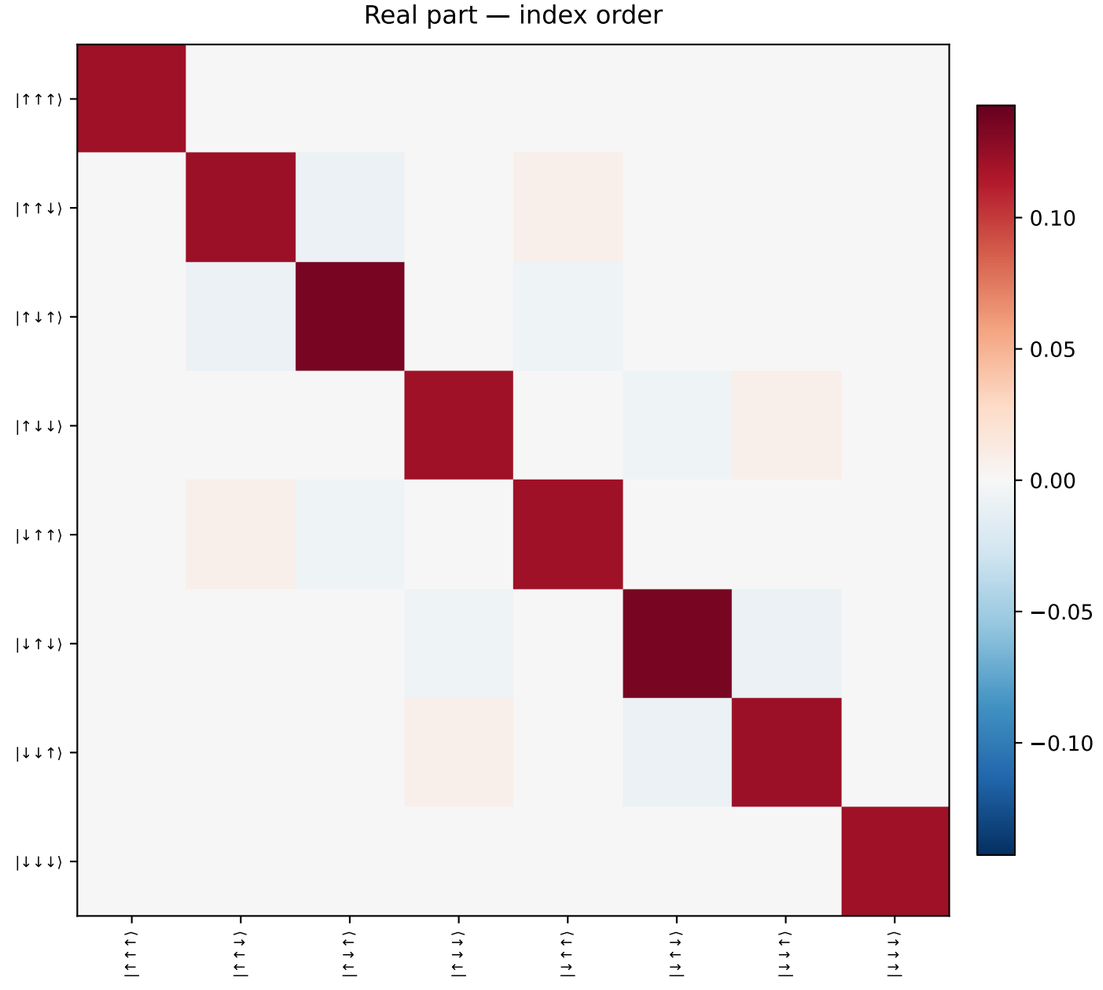
gamma = Tr(rho^2) = 0.49978192

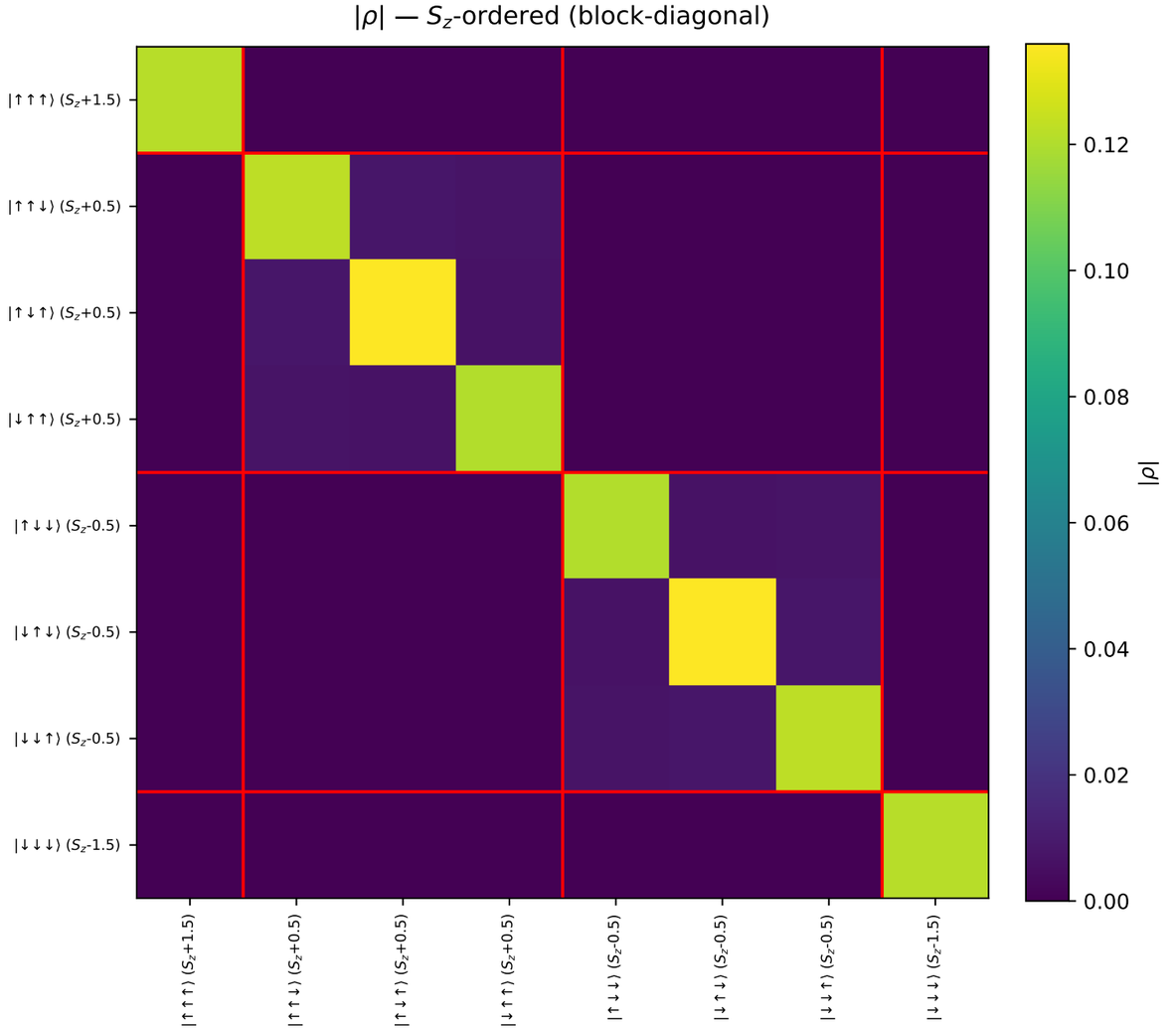
A	B	Tr(rho_A^2)

STATUS: MIXED STATE

Tr(rho^2) < 1. Pure-state C_GME not applicable. See convex-roof result below.

3-Body RDM (Sites 2,3,5)





Trace=1.000000 Purity=0.125967 Entropy=2.075664 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 ($\Rightarrow S_z$ conserved)

Analysis Summary - 3-Body RDM (Sites 2,3,5)

RDM Metrics
==========

Trace(rho)	: 1.0000000000
Purity Tr(rho^2)	: 0.1259673001
Von Neumann entropy	: 2.0756636393
Eigenvalue min	: 1.140523e-01
Eigenvalue max	: 1.433427e-01
Imaginary part	: < 1e-6 (not shown)

GME Analysis
==========

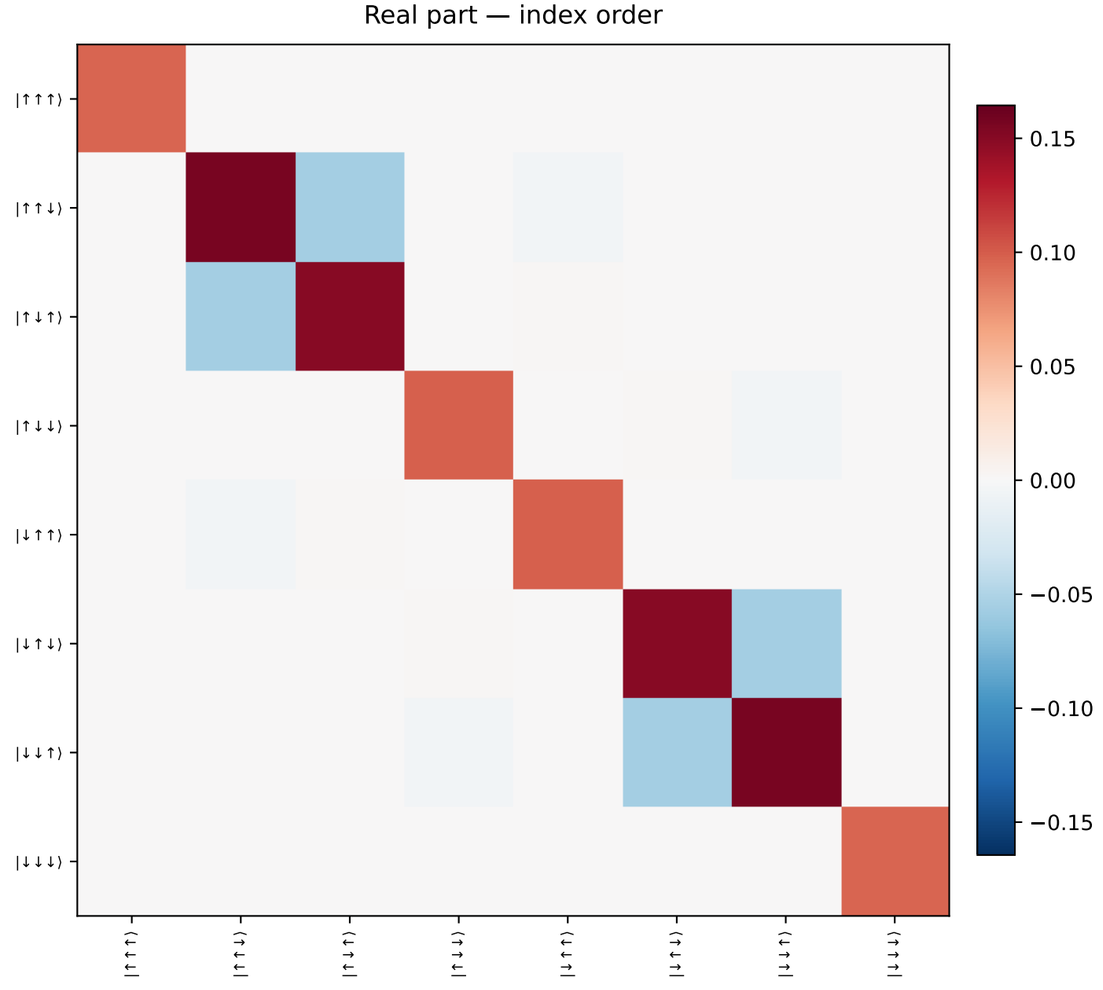
gamma = Tr(rho^2) = 0.12596730

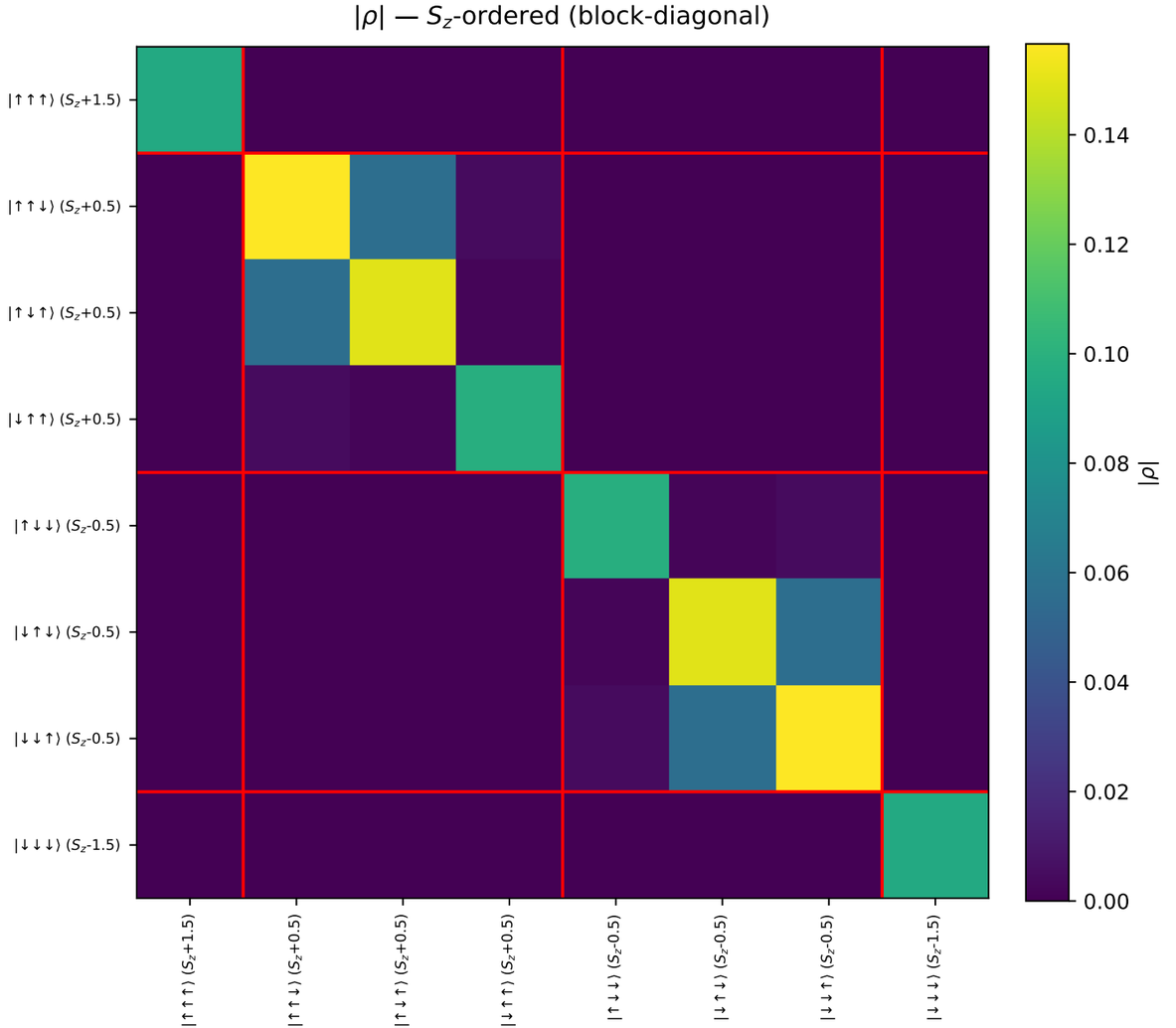
A	B	Tr(rho_A^2)

STATUS: MIXED STATE

Tr(rho^2) < 1. Pure-state C_GME not applicable. See convex-roof result below.

3-Body RDM (Sites 2,4,5)





Trace=1.000000 Purity=0.144255 Entropy=2.010634 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 ($\Rightarrow S_z$ conserved)

Analysis Summary - 3-Body RDM (Sites 2,4,5)

RDM Metrics
==========

Trace(rho)	: 1.0000000000
Purity Tr(rho^2)	: 0.1442552519
Von Neumann entropy	: 2.0106344236
Eigenvalue min	: 9.531950e-02
Eigenvalue max	: 2.099249e-01
Imaginary part	: < 1e-6 (not shown)

GME Analysis
==========

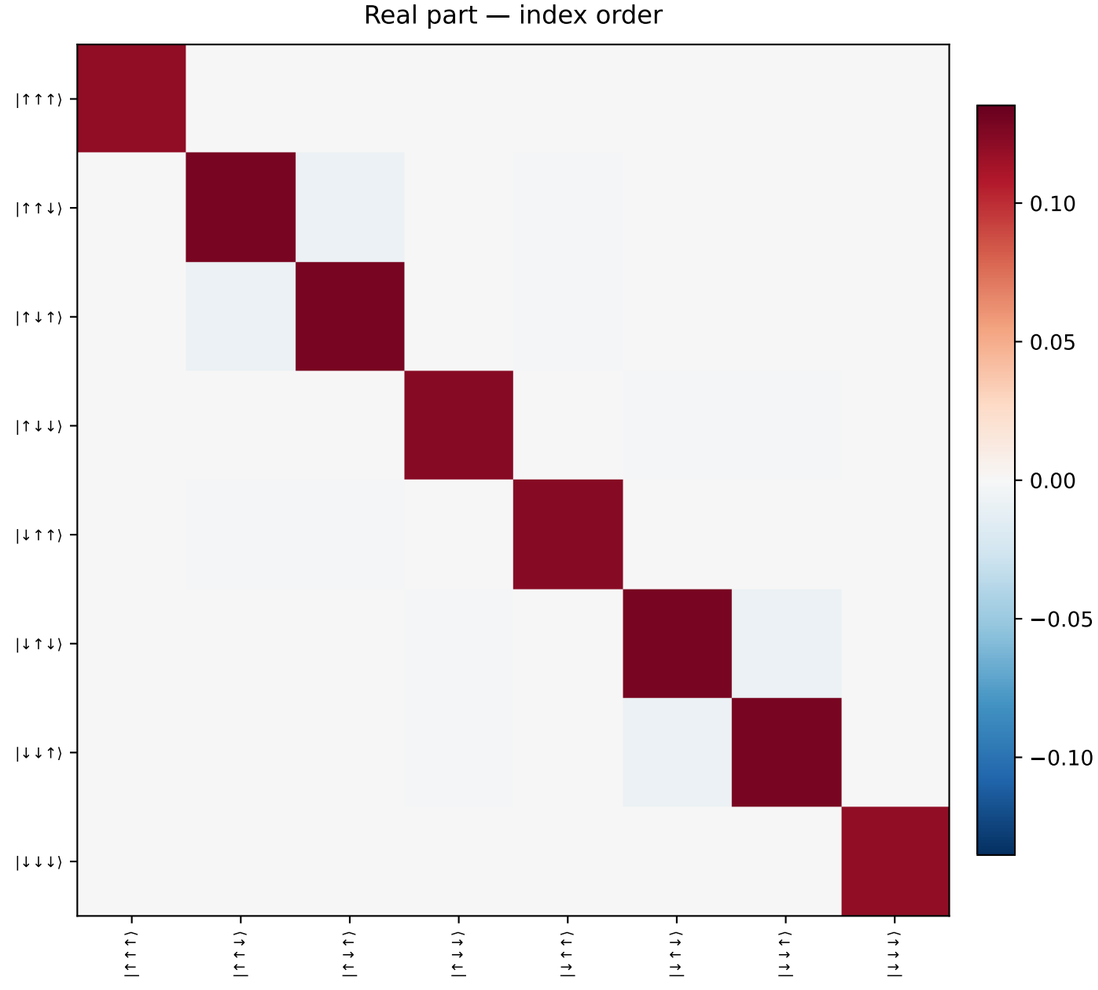
gamma = Tr(rho^2) = 0.14425525

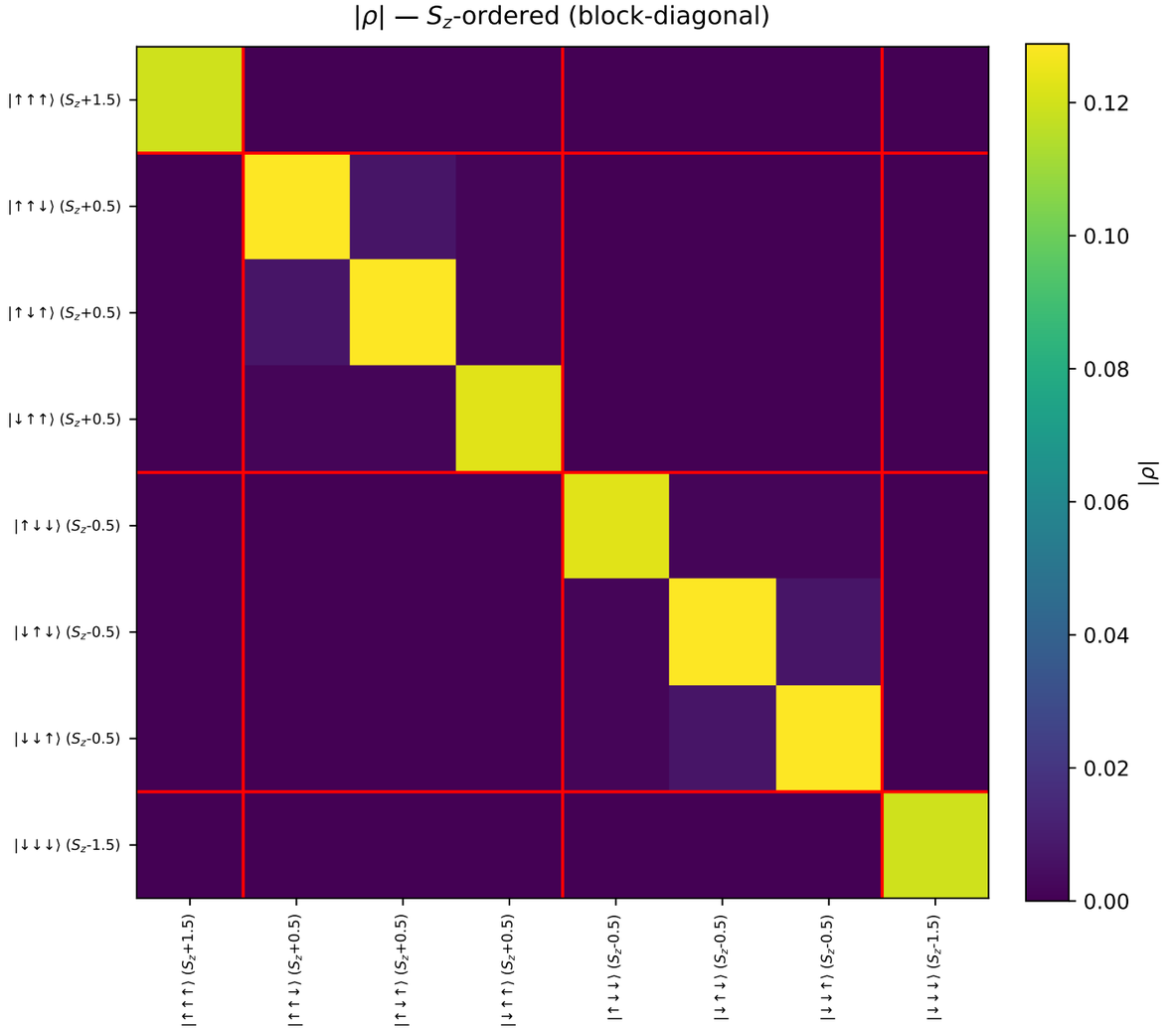
A	B	Tr(rho_A^2)

STATUS: MIXED STATE

Tr(rho^2) < 1. Pure-state C_GME not applicable. See convex-roof result below.

3-Body RDM (Sites 3,4,5)





Trace=1.000000 Purity=0.125368 Entropy=2.077991 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 ($\Rightarrow S_z$ conserved)

Analysis Summary - 3-Body RDM (Sites 3,4,5)

RDM Metrics
==========

Trace(rho)	: 1.0000000000
Purity Tr(rho^2)	: 0.1253680220
Von Neumann entropy	: 2.0779912025
Eigenvalue min	: 1.195895e-01
Eigenvalue max	: 1.361948e-01
Imaginary part	: < 1e-6 (not shown)

GME Analysis
==========

gamma = Tr(rho^2) = 0.12536802

A	B	Tr(rho_A^2)

STATUS: MIXED STATE

Tr(rho^2) < 1. Pure-state C_GME not applicable. See convex-roof result below.